MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

ULTRAVIOLET LIGHT ERASABLE PROGRAMMABLE READ ONLY MEMORY (UV EPROM) MODULE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Ultraviolet Light Erasable Programmable Read Only Memory (UV EPROM) module.

1.2 Classification. This specification covers two classes of modules. Class I and class II are as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.1.1 Application requirements. This specification establishes the general requirements and test procedures for an Ultra Violet Erasable Programable Read Only Memory (UV EPROM) module. When a Standard Electronic Module (SEM) user determines a requirement for a UV EPROM module, the user shall prepare a Design Approval Request (DAR) and submit the DAR to the SEM Design Review Activity (SEM-DRA) in accordance with MIL-STD-1378. The DAR shall contain, as a minimum, the module key code, the firmware pattern, and detailed module marking.

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 5.0731 failures per million operating hours for class I or 16.6 failures per million operating hours for class II when predicted in accordance with MIL-HDBK-217. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I, and the user's DAR and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

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3.6.1 Key code. The module key code shall be as specified in the user's DAR.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 90 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 6/	Ref test table	Test type 1/-4/	25 ± 5°C 5/		Class I 5/ 0 ± 5°C 60 ∓ 5°C		Class II 5/ -55 ± 5°C 85 ∓ 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation									
	initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	S	1	INF	1	INF	1	INF	MOhm
A2-A24	Isolation									
	initial	II	X	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	X	1	INF	1	INF	1	INF	MOhm
B1-B8	Continuity	III	X	0	1	0	1	0	1	Ohm
C1-C17	Ibd	IV	S	-20	20	-20	20	-20	20	uA
C18-C34	Ibd	IV	S	0	100	0	100	0	100	uA
C35	Ibd	IV	S	-113	113	-113	113	-113	113	uA
D1-D4	Ios	V	S	-375	-100	-375	-100	-375	-100	mA
D5-D12	Ios	V	S	-150	-60	-150	-60	-150	-60	mA
E	Vcc stress	VI	S	----------	Module shall pass all		remaining tests		-------	
F1-F34	Vic	VII	S	-1.2	-0.2	-1.2	-0.2	-1.2	-0.2	V
G1-G17	Iil	VIII	X	-20	20	-20	20	-20	20	uA
G18-G26	Iil	VIII	X	-1000	100	-1000	100	-1000	100	uA
G27-G34	Iil	VIII	X	-600	60	-600	60	-600	60	uA
G35	Iil	VIII	X	-5500	-4500	-5500	-4500	-5500	-4500	uA
G36-G51	Iil	VIII	X	-10	10	-10	10	-10	10	uA
H1-H17	Iih	IX	X	-20	20	-20	20	-20	20	uA
H18-H34	Iih	IX	X	-2	20	-2	20	-2	20	uA
H35	Iih	IX	X	-3080	-2520	-3080	-2520	-3080	-2520	uA
H36-H51	Iih	IX	X	-10	10	-10	10	-10	10	uA
J1-J16	Iozh	X	X	-10	10	-10	10	-10	10	uA
J17-J20	Iozh	X	X	-5	50	-5	50	-5	50	uA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-4/	25 ± 5°C 5/		Class I 5/ 0 ± 5°C 60 ± 5°C		Class II 5/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
K1-K16	Iozl	XI	X	-10	10	-10	10	-10	10	uA
K17-K20	Iozl	XI	X	-50	5	-50	5	-50	5	uA
L1-L16	Voh	XII	X	3.7	4.5	3.7	4.5	3.7	4.5	V
L17-L20	Voh	XII	X	2.4	4.5	2.4	4.5	2.4	4.5	V
L21-L28	Voh	XII	X	2.5	4.5	2.5	4.5	2.5	4.5	V
M1-M16	Vol	XIII	X	0.0	0.5	0.0	0.5	0.0	0.5	V
M17-M20	Vol	XIII	X	0.0	0.55	0.0	0.55	0.0	0.55	V
M21-M28	Vol	XIII	X	0.0	0.5	0.0	0.5	0.0	0.5	V
N1	Icc	XIV	X	0	96	0	96	0	96	mA
N2	Idd	XIV	X	0	100	0	100	0	100	mA
N3	Ioc	XIV	X	0	96	0	96	0	96	mA
N4	Idd	XIV	X	0	200	0	200	0	200	uA
P1	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
R1-R20	Enable time (tPZH)	XVIII	S	0	40	0	40	0	40	nS
R21-R40	Enable time (tPZL)	XVIII	S	0	40	0	40	0	40	nS
R41-R60	Disable time (tPHZ)	XVIII	S	0	40	0	40	0	40	nS
R61-R80	Disable time (tPLZ)	XVIII	S	0	40	0	40	0	40	nS
S1-S19	Cin	XIX	Q	2	30	2	30	2	30	pF
S20-S35	Cin	XIX	Q	2	15	2	15	2	15	pF
S36-S51	Cout	XIX	Q	2	20	2	20	2	20	pF
S52-S63	Cout	XIX	Q	2	35	2	35	2	35	pF

- See footnotes at end of table.

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TABLE I Module electrical requirements - Continued.

Test cond	Parameter 6/	Ref test table	Test type 1/-4/	25 ± 5°C 5/		Class I 5/ 0 ± 5°C 60 ± 5°C		Class II 5/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
T1-T16	Output crosstalk high level	XX	Q	0	550	0	550	0	550	mV
T17-T28	Output crosstalk high level	XX	Q	0	400	0	400	0	400	mV
T29-T44	Output crosstalk low level	XX	Q	0	550	0	550	0	550	mV
T45-T56	Output crosstalk low level	XX	Q	0	400	0	400	0	400	mV
U	PSTA	XXI	Q	0	400	0	400	0	400	mV
V1	PSFC	XXII	Q	3.71	6.89	3.71	6.89	3.71	6.89	uF
V2	PSFC	XXII	Q	3.43	6.37	3.43	6.37	3.43	6.37	uF
W1-W24	Propagation delay time	XXIII	S	0	34	0	34	0	34	nS
X	Erasing	XXIV	Q	----------See table XXIV----------						
Y	Programming	XXV	Q	----------See table XXV----------						
Z	Margin	XXVI	Q	----------See table XXVI----------						

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ "X" parameters, except test conditions A2-A24 and B1-B8, shall be 100 percent tested when modules are reprogramed.

5/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

6/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/458-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.8.1 Reprogramed module test. All reprogramed modules shall be tested in accordance with footnote 4 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

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4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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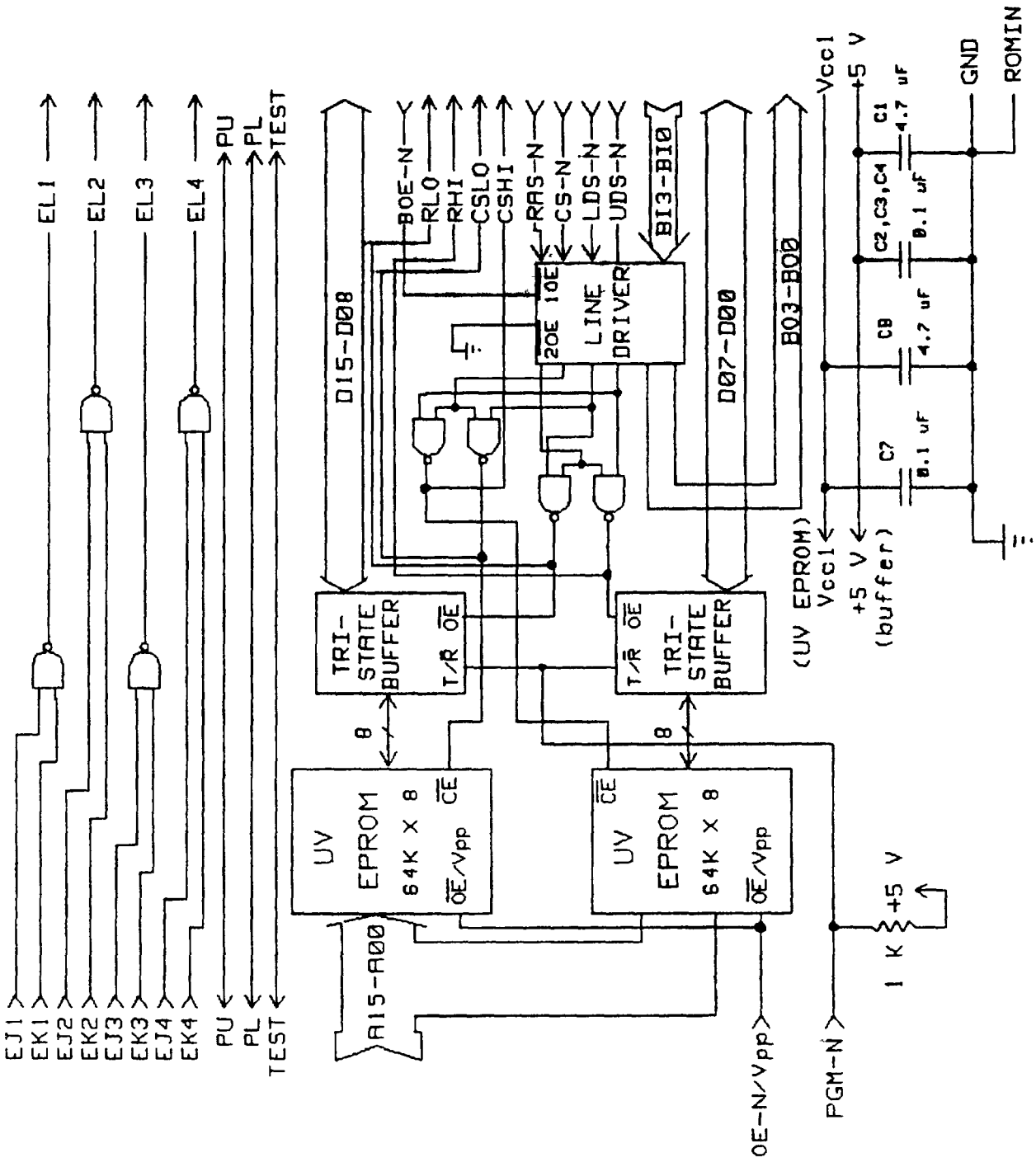


FIGURE 1. Functional block diagram.

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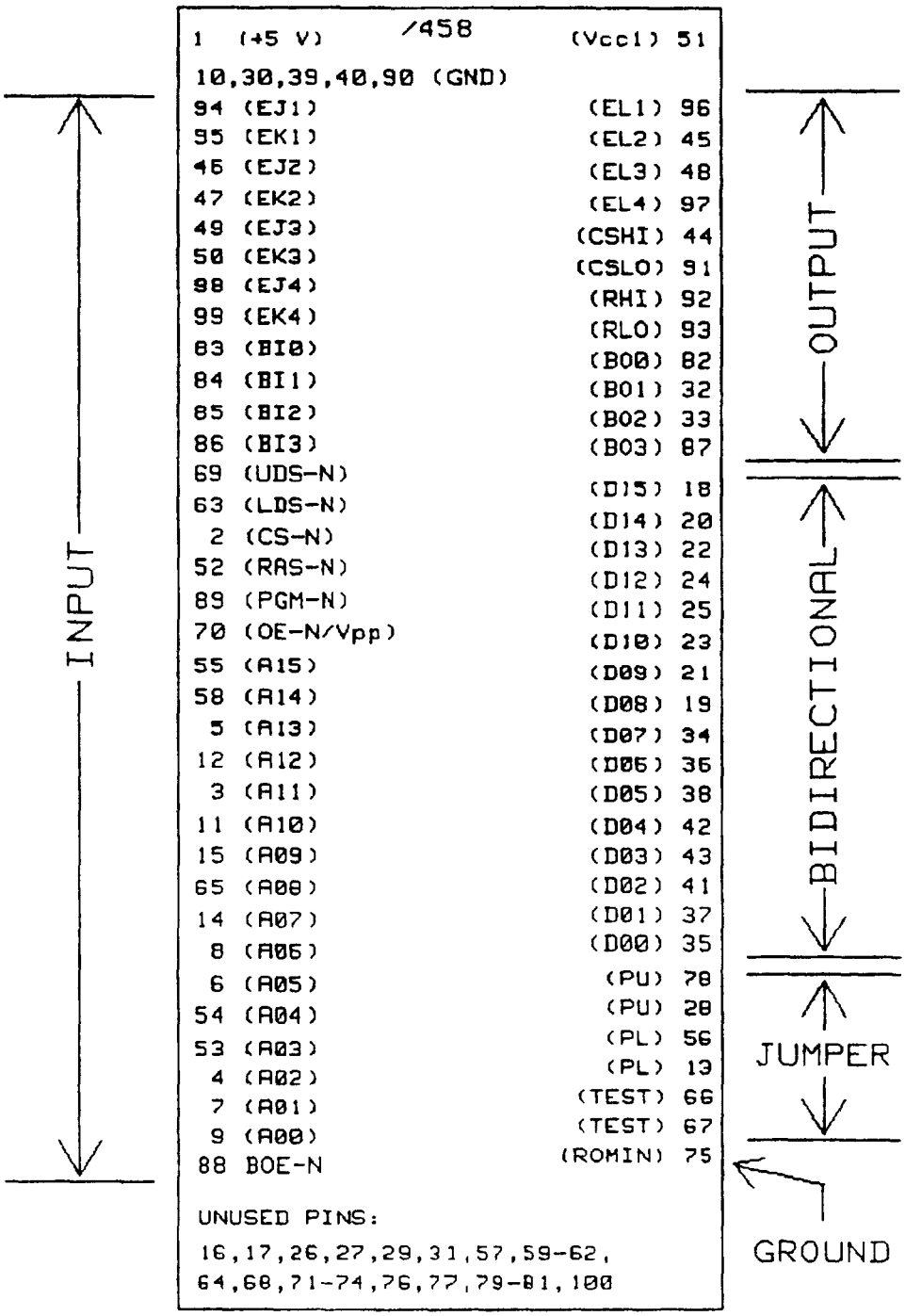


FIGURE 2. Pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through AC shall be performed in any order with the following exceptions. Test conditions A1, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A1, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A1, C, D, E and F where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

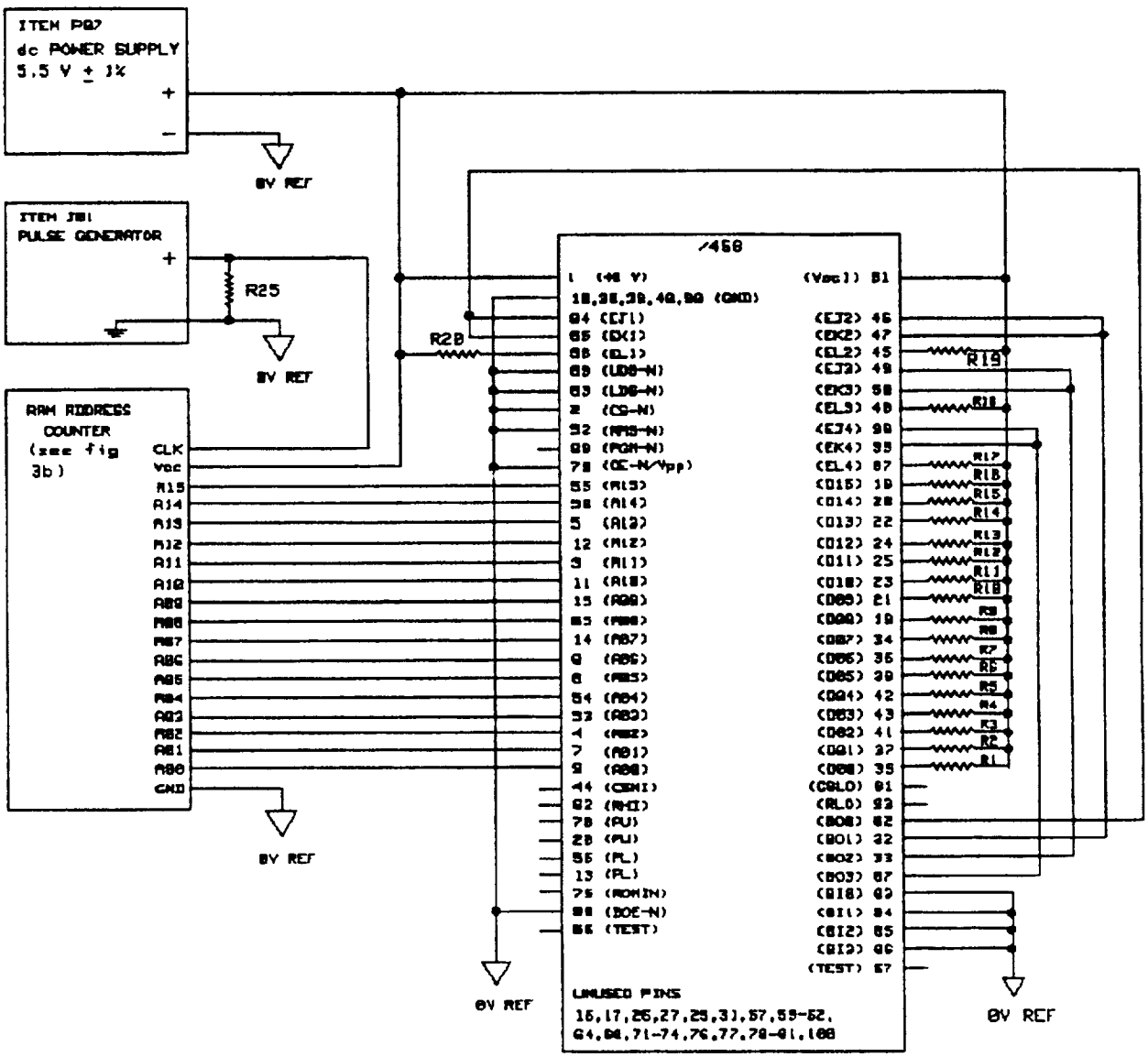
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AC in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through T, and V through Z shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figures 3 and 4.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition Y, shall be performed using the initial limits at 25 ± 5°C ambient.

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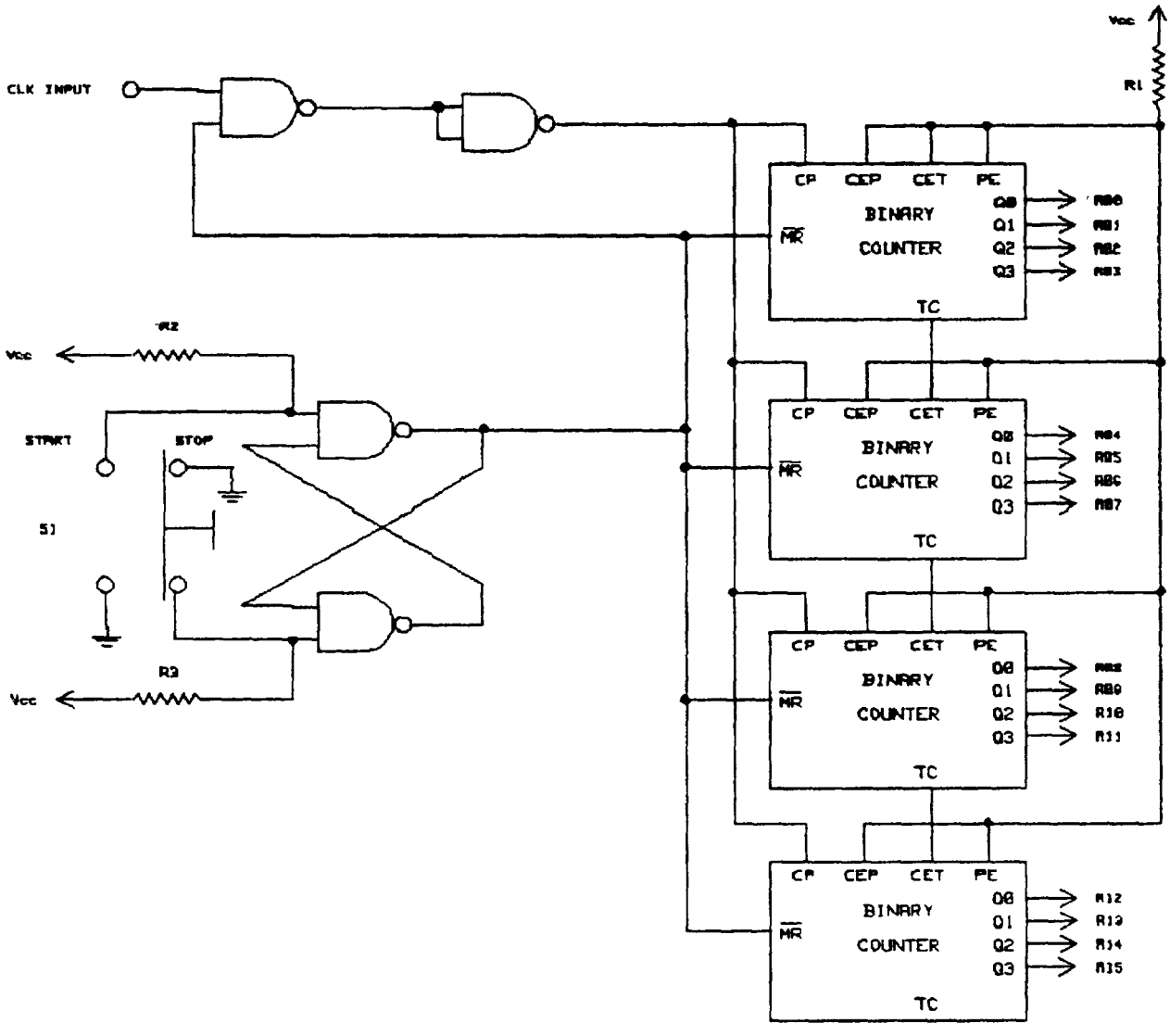
NOTES:

1. R1-R16 = 500 Ohms ± 1 percent, 1/8 W; R17-R20 = 255 Ohms ± 1 percent, 1/8 W; R25 = 51 Ohms ± 5 percent, 1 W.

2. An address shall be selected that results in a minimum of eight data outputs being at a logic level low.

FIGURE 3a. Operating temperature and life test circuit.

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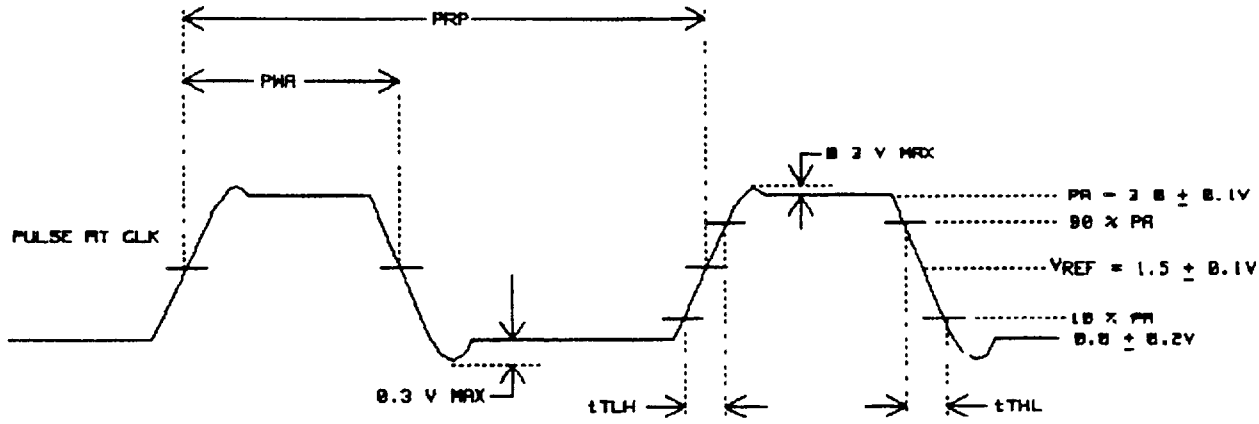


NOTES:

1. R1 = R2 = R3 = 2.2 KOhms ± 10 percent, 1/8 W.
2. S1 = push button switch.

FIGURE 3b. RAM address counter circuit.

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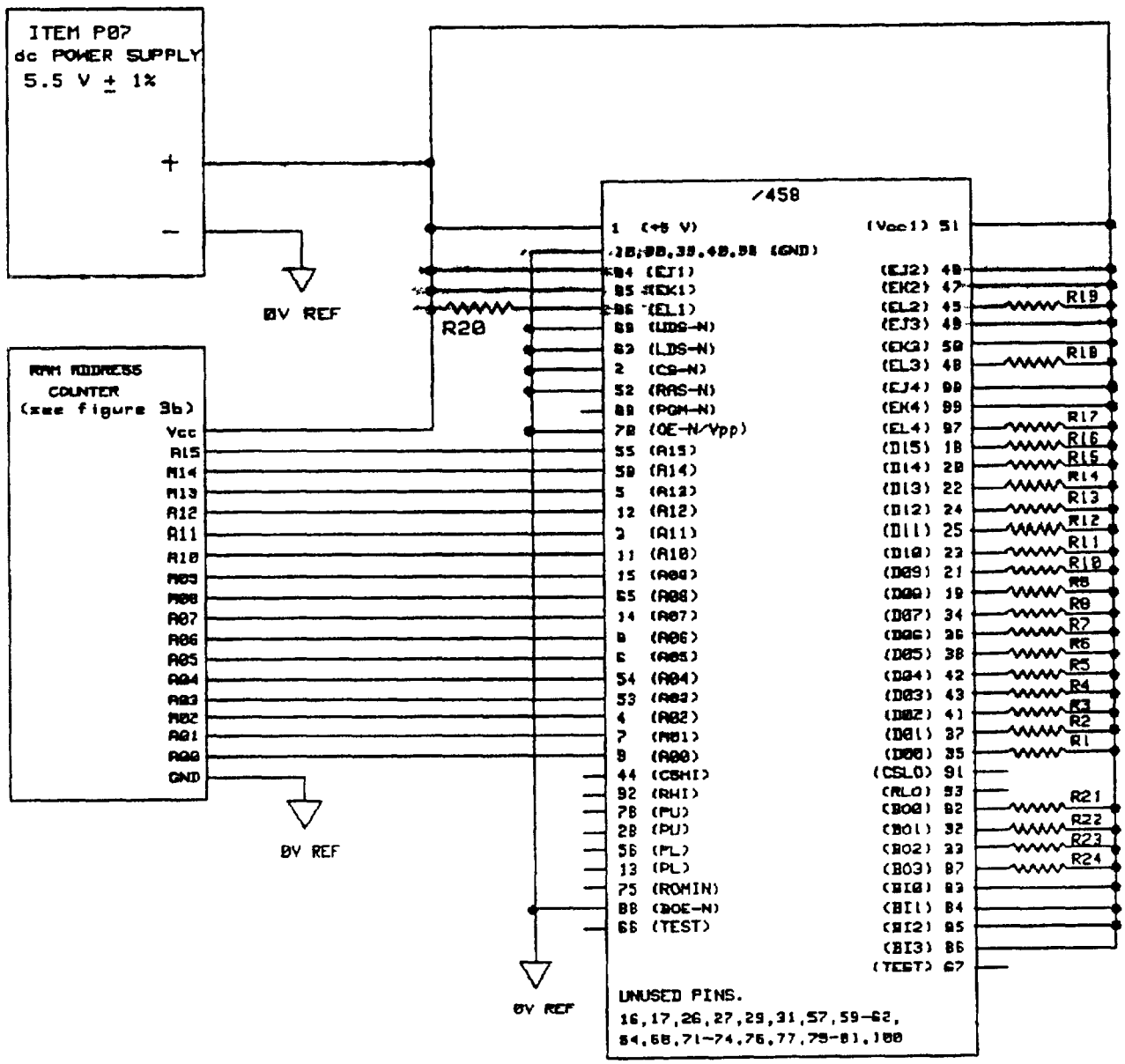


NOTE:

1. Input pulse parameters: tTLH = tTHL = 20 ± 10 nS, PWA = 100 ± 10 nS, PRP = 300 ± 10 nS.

FIGURE 4. Operating temperature and life test waveforms.

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NOTES:

1. R1-R16 = 500 Ohms ± 1 percent, 1/8 W; R17-R20 = 255 Ohms ± 1 percent, 1/8 W; R21-R24 = 432 Ohms ± 1/8 W.

2. An address shall be selected that results in a minimum of eight data outputs being at a logic level low.

FIGURE 5. Component temperature test circuit.

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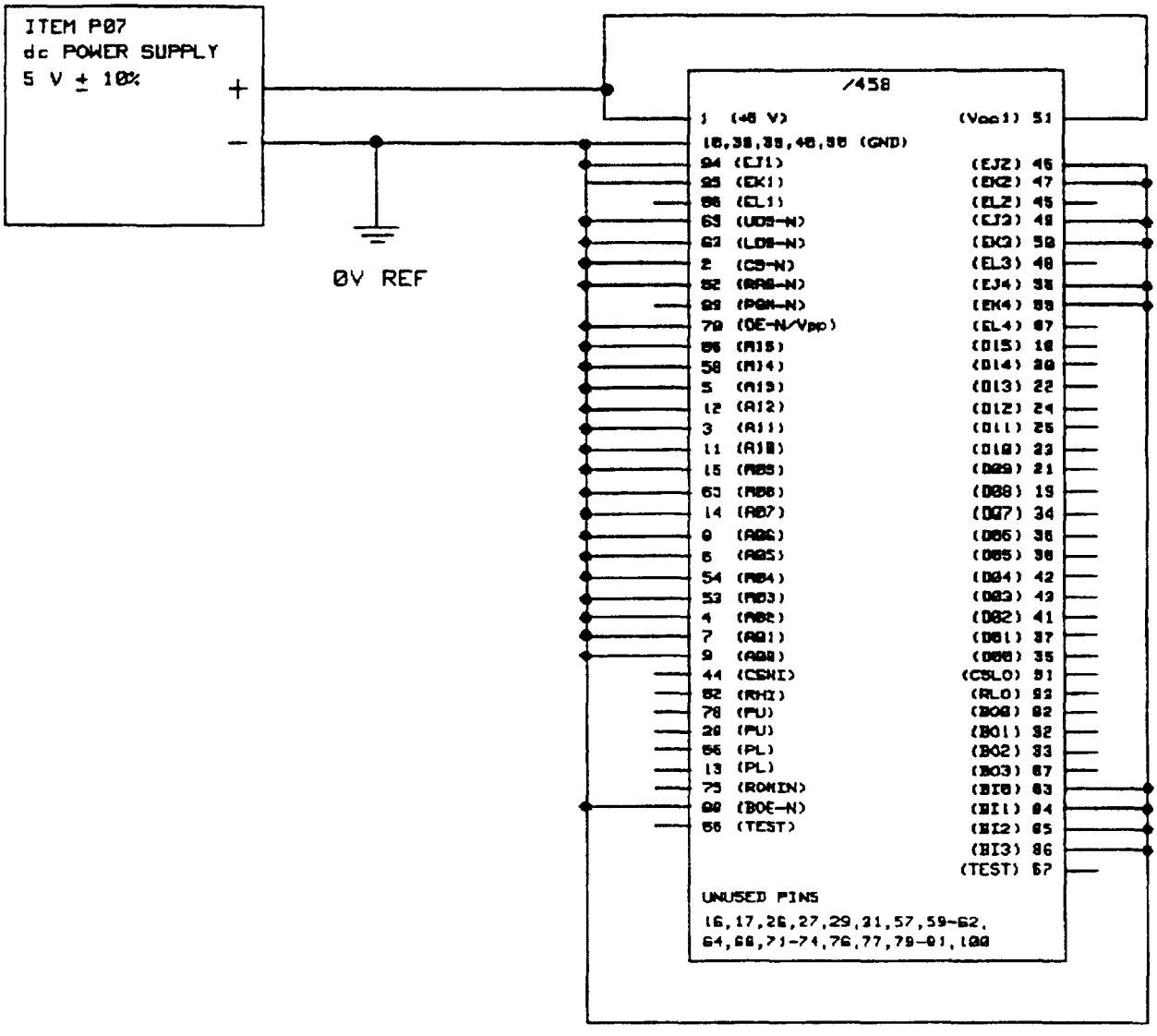


FIGURE 6. Vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4. Following the life test and while still at the maximum class temperature, electrical tests B, G through T, and V through Z, shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through T, and V through Z, shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

4.6 Reprogramed testing. Reprogramed module tests shall be performed when modules are reprogramed.

4.6.1 Reprogramed tests. Test conditions G through M and R in table I shall be performed when a module has been reprogramed. Tests shall be performed at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Din = input data.
(d) Dout = output data.
(e) Ibd = input breakdown current.
(f) Icc = power supply current for logic.
(g) Idd = power supply current for UV EPROM.
(h) Iih = input high level current.
(i) Iil = input low level current.
(j) Iin = input forcing current.
(k) Ioh = output high level current.
(l) Iol = output low level current.
(m) Ios = output short circuit current.
(n) Iozh = output high level leakage current.
(o) Iozl = output low level leakage current.
(p) NA = not applicable.
(q) PSFC = power supply filter capacitance.
(r) PSTA = power supply transient amplitude.
(s) Vic = input clamp diode voltage.
(t) Vih = input high level voltage.
(u) Vil = input low level voltage.
(v) Vin = input forcing voltage.
(w) Voh = output high level voltage.
(x) Vol = output low level voltage.
(y) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 3.6 W (max); 2.066 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
A15-A00	-1.2	0.8	2.0	-0.02	0.02	0.02	30.0
D15-D00	-1.2	0.8	2.0	-0.01	0.01	NA	15.0
RAS-N	-1.2	0.8	2.0	-1.0	0.02	0.1	15.0
CS-N	-1.2	0.8	2.0	-1.0	0.02	0.1	15.0
LDS-N	-1.2	0.8	2.0	-1.0	0.02	0.1	15.0
UDS-N	-1.2	0.8	2.0	-1.0	0.02	0.1	15.0
BI3-BI0	-1.2	0.8	2.0	-1.0	0.02	0.1	15.0
BOE-N	-1.2	0.8	2.0	-1.0	0.02	0.1	30.0
EJ4-EJ1	-1.2	0.8	2.0	-0.6	0.02	0.1	15.0
EK4-EK1	-1.2	0.8	2.0	-0.6	0.02	0.1	15.0
OE-N/Vpp	-1.2	0.8	2.0	-0.02	0.02	0.02	30.0
PGM-N	-1.2	0.8	2.0	-7.5	0.32	0.113	30.0

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh min (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
D15-D00	3.7	0.5	-24.0	24.0	10.0	-10.0	NA	20.0
CSHI	2.5	0.5	-1.0	20.0	NA	NA	-60.0	35.0
CSLO	2.5	0.5	-1.0	20.0	NA	NA	-60.0	35.0
RHI	2.5	0.5	-1.0	20.0	NA	NA	-60.0	35.0
RLO	2.5	0.5	-1.0	20.0	NA	NA	-60.0	35.0
BO3-BO0	2.4	0.55	-3.0	48.0	50.0	-50.0	-100.0	20.0
EL4-EL1	2.5	0.5	-1.0	20.0	NA	NA	-60.0	20.0

6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The UV EPROM module is organized as two 64K word by 8 bit memories that may be operated in series or parallel from essentially the same address inputs. Therefore, the module can be configured as either 128K word by 8 bit or 64K word by 16 bit memory with a total Read Only Memory (ROM) that is 1,048,576 bits in capacity. The data lines are buffered to ensure compatibility when used in systems with large bus loads.

6.4.1 Accessing memory. The UV EPROM module provides the ability to access memory from the UV EPROMs in either the 64K word by 16 bit or 128K word by 8 bit configuration within 174.5 nS. The 64K words by 8 bits is the configuration when two UV EPROMs are read at the same time. The 128K words by 8 bits is the configuration when two UV EPROMs are read at different times.

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The address lines A15-A00 must be stable 180 nS before the data can be stable at the outputs of the transceivers. Signals CSHI and CSLO are the chip enables of the UV EPROMs and must be low 150 nS before the data is stable at the module output. CSHI and CSLO go low between 5 and 16 nS after LDS-N, UDS-N and CS-N go low. RHI and RLO are the output enables of the transceivers and must be low between 1 and 10 nS before the data is stable at the module output. Signals RHI and RLO go low between 5 and 16 nS after LDS-N, UDS-N and RAS-N go low. Signal OE-N/Vpp is the output enable of the UV EPROMs and must be low 70 ns before the data is stable at the module output. Signal PGM-N is the bidirectional control pin of the module. When signal PGM-N is pulled low, the signals D15-D00 become inputs. At all other times this PGM-N signal is active high (internal pullup resistor), and the module is always in an output state. The access time is not dependent on whether the 64K word by 16 bit or the 128K word by 8 bit configuration is used. The module is configured for 64K words by 8 bits by enabling simultaneously both LDS-N and UDS-N. The module is configured for 128K words by 8 bits by enabling UDS-N or LDS-N separately.

6.4.2 Standby. The complete standby mode is defined when the two signals CSHI and CSLO are active high and the program mode is not defined. When this condition is met, the power supply current for the two UV EPROMs will drop from 100 mA to 200 uA. The information below defines the different mode operations of the module.

Vcc1	UDS-N	LDS-N	RAS-N	CS-N	OE-N/Vpp	PGM-N	D07-D00	D15-D08	Operation mode
5.0 V	L	L	L	L	L	H	Dout	Dout	64K x 16
5.0 V	H	L	L	L	L	H	Dout	Z	128K x 8
5.0 V	L	H	L	L	L	H	Z	Dout	128K x 8
5.0 V	L	L	L	L	12.5 V	L	Din	Din	64K x 16
5.0 V	H	L	L	L	12.5 V	L	Din	Z	128K x 8
5.0 V	L	H	L	L	12.5 V	L	Z	Din	128K x 8
5.0 V	X	X	H	H	X	X	Z	Z	Standby
5.0 V	L	L	L	L	L	H	Dout	Dout	Manufacturer's code

6.4.3 Manufacturer code. A code is permanently burned in ROM on the UV EPROM. This code identifies the UV EPROM maker and the device type.

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6.4.4 Miscellaneous. This module has additional capability in the form of four independent, two input NAND gates and one quad independent inverting buffer. Also, there are three pairs of jumper connections and an additional ground pin (see 6.4.4.1, 6.4.4.2, 6.4.4.3 and figure 1).

6.4.4.1 NAND gates. Input signals EJ1-EJ4 and EK1-EK4 are NANDed to produce the output signals EL1-EL4.

6.4.4.2 Line driver. The four output signals BO0-BO3 are the inverting signals of BI0-BI3. These signals can be enabled by bringing the input signal BOE-N to low.

6.4.4.3 Jumpers and ground. There are three pairs of pins (PU, PL and TEST) which are jumper connections. The pin ROMIN is tied to ground.

6.4.5 Description of I/O pins.

Signal name	Pin number	I/O	Functional description
A15	55	I	These signals are address bus bits. They are the direct input signals of the two UV EPROMs.
A14	58	I	
A13	5	I	
A12	12	I	
A11	3	I	
A10	11	I	
A09	15	I	
A08	65	I	
A07	14	I	
A06	8	I	
A05	6	I	
A04	54	I	
A03	53	I	
A02	4	I	
A01	7	I	
A00	9	I	
OE-N/Vpp	70	I	This signal is both an output enable and a programming enable for the UV EPROMs.

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
UDS-N	69	I	This signal is an upper data strobe. It is also an input to the inverting buffer.
LDS-N	63	I	This signal is a lower data strobe. It is also an input to the inverting buffer.
CS-N	2	I	This signal is an indirect chip enable for the UV EPROMs. It is also an input to the inverting buffer.
RAS-N	52	I	This signal is an indirect out control signal of the data transceivers. It is also the input to the inverting buffer.
BI0 BI1 BI2 BI3	83 84 85 86	I I I I	These four signals are the inputs of the inverting buffer.
BOE-N	88	I	This signal is the output enable for the output signals BO0-BO3.
EJ1 EJ2 EJ3 EJ4 EK1 EK2 EK3 EK4	94 46 49 98 95 47 50 99	I I I I I I I I	These signals are the input signals of the four independent, two input NAND gates.
PGM-N	89	I	This signal is the directional control pin of the module's bidirectional signals D15-D00. It is also the directional control pin of a set of data

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
			transceivers. When it is active low, the data bus becomes input. Otherwise, at all other times it is active high, because of the internal pullup resistor on the module.
BO0 BO1 BO2 BO3	82 32 33 87	O O O O	These signals are the outputs of the independent inverting buffer. These signals are enabled by the input signal BOE-N.
EL1 EL2 EL3 EL4	96 45 48 97	O O O O	These signals are the outputs of the four independent, two input NAND gates.
D00 D01 D02 D03 D04 D05 D06 D07 D08 D09 D10 D11 D12 D13 D14 D15	35 37 41 43 42 38 36 34 19 21 23 25 24 22 20 18	I/O I/O I/O I/O I/O I/O I/O I/O I/O I/O I/O I/O I/O I/O I/O I/O	This is the bidirectional data bus of the module. They are also the bidirectional signals of a set of transceivers. These signals' flow directions are controlled by the input signal PGM-N.
CSHI	44	O	This signal is the chip select of the upper UV EPROMs. It is also the output of a NAND gate. This signal is produced by NANDing the two input signals, CS-N and UDS-N.

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
CSLO	91	O	This signal is the chip select of the lower UV EPROMs. It is also the output of a NAND gate. This signal is produced by NANDing the two input signals, CS-N and LDS-N.
RHI	92	O	This signal is the output enable of the upper data transceiver. It is also the output signal of a NAND gate. This signal is produced by NANDing the two input signals, UDS-N and RAS-N.
RLO	93	O	This signal is the output enable of the lower data transceiver. It is also the output signal of a NAND gate. This signal is NANDed by two input signals, LDS-N and RAS-N.

6.5 Module programming. The module programming is printed in the appendix.

6.6 Erasability. The UV EPROM module shall be erasable by exposure to a high intensity ultraviolet light at a wavelength of 2537 angstroms. A complete erasure is assured by placing the UV EPROM window within one inch and directly underneath an ultraviolet lamp with a wavelength of 2537 angstroms, intensity of 12,000 uW/cm*cm for a period of 20 minutes. After erasure, all address locations shall be tested to read active high. Minimum erase time for lamps at other intensity ratings can be calculated from the minimum integrated erasure dose of 15 W-s/cm*cm. To prevent unintentional erasure, an opaque label is recommended to cover the clear windows of the two UV EPROMs.

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6.7 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Ibd test - table IV.

(d) Ios test - table V.

(e) Vcc stress test - table VI.

(f) Vic test - table VII.

(g) Iil test - table VIII.

(h) Iih test - table IX.

(i) Iozh test - table X.

(j) Iozl test - table XI.

(k) Voh test - table XII, figure 7.

(l) Vol test - table XIII, figure 7.

(m) Power supply current test - table XIV.

(n) Functional test - table XV.

(o) Dynamic test - tables XVI and XVII, figures 7 and 8.

(p) Disable/Enable time test - table XVIII, figure 9.

(q) Capacitance test - table XIX.

(r) Output crosstalk test - table XX, figures 7, 10 and 11.

(s) PSTA - table XXI, figures 12 and 13.

(t) PSFC test - table XXII.

(u) Propagation delay test - table XXIII, figures 7 and 14.

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(v) Erasing test - table XXIV.

(w) Programming test - table XXV.

(x) Margin test - table XXVI.

6.8 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N748)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units
A1	Isolation, frame	All pins	Frame	10	INF	MOhm
A2	Isolation, used to unused	All used pins	All unused pins	10	INF	MOhm
A3	Isolation, unused to unused	16	17,26,27, 29,31,57, 59,60,61, 62,64,68, 71,72,73, 74,76,77, 79,80,81, 100	10	INF	MOhm
A4	Isolation, unused to unused	17	26,27,29, 31,57,59, 60,61,62, 64,68,71, 72,73,74, 76,77,79, 80,81,100			
A5	Isolation, unused to unused	26	27,29,31, 57,59,60, 61,62,64, 68,71,72, 73,74,76, 77,79,80, 81,100			
A6	Isolation, unused to unused	27	29,31,57, 59,60,61, 62,64,68, 71,72,73, 74,76,77, 79,80,81, 100	10	INF	MOhm

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A7	Isolation, unused to unused	29	31,57,59, 60,61,62, 64,68,71, 72,73,74, 76,77,79, 80,81,100	10	INF	MOhm	1-7
A8	Isolation, unused to unused	31	57,59,60, 61,62,64, 68,71,72, 73,74,76, 77,79,80, 81,100				
A9	Isolation, unused to unused	57	59,60,61, 62,64,68, 71,72,73, 74,76,77, 79,80,81, 100				
A10	Isolation, unused to unused	59	60,61,62, 64,68,71, 72,73,74, 76,77,79, 80,81,100				
A11	Isolation, unused to unused	60	61,62,64, 68,71,72, 73,74,76, 77,79,80, 81,100				
A12	Isolation, unused to unused	61	62,64,68, 71,72,73, 74,76,77, 79,80,81, 100	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A13	Isolation, unused to unused	62	64,68,71, 72,73,74, 76,77,79, 80,81,100	10	INF	MOhm	1-7
A14	Isolation, unused to unused	64	68,71,72, 73,74,76, 77,79,80, 81,100				
A15	Isolation, unused to unused	68	71,72,73, 74,76,77, 79,80,81, 100				
A16	Isolation, unused to unused	71	72,73,74, 76,77,79, 80,81,100				
A17	Isolation, unused to unused	72	73,74,76, 77,79,80, 81,100				
A18	Isolation, unused to unused	73	74,76,77, 79,80,81, 100				
A19	Isolation, unused to unused	74	76,77,79, 80,81,100				
A20	Isolation, unused to unused	76	77,79,80, 81,100				
A21	Isolation, unused to unused	77	79,80,81, 100	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A22	Isolation, unused to unused	79	80,81,100	10	INF	MOhm	1-7
A23	Isolation, unused to unused	80	81,100				
A24	Isolation, unused to unused	81	100	10	INF	MOhm	1-7

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V 30 V	± 10 percent ± 10 percent	A1 A2-A24
Isolation measurement (see note 6)	NA NA	± 500 KOhm ± 50 KOhm	A1-A24 A1-A24

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

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NOTES: - Continued.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	30	0	1	Ohm	1-4
B2	Continuity, GND	10	39				
B3	Continuity, GND	10	40				
B4	Continuity, GND	10	90				
B5	Continuity, ROMIN	10	75				
B6	Continuity, PU	78	28				
B7	Continuity, PL	56	13				
B8	Continuity, TEST	66	67	0	1	Ohm	1-4

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 50 MOhm	B1-B8

2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

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TABLE IV. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
C1	A15	55	5.5	-20	20	uA	1-3
C2	A14	58					
C3	A13	5					
C4	A12	12					
C5	A11	3					
C6	A10	11					
C7	A09	15					
C8	A08	65					
C9	A07	14					
C10	A06	8					
C11	A05	6					
C12	A04	54					
C13	A03	53					
C14	A02	4					
C15	A01	7					
C16	A00	9					
C17	OE-N/Vpp	70	5.5	-20	20	uA	1-3
C18	UDS-N	69	5.5	0	100	uA	1-3
C19	LDS-N	63					
C20	CS-N	2	5.5	0	100	uA	1-3

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TABLE IV. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
C21	RAS-N	52	5.5	0	100	uA	1-3
C22	BI0	83					
C23	BI1	84					
C24	BI2	85					
C25	BI3	86					
C26	BOE-N	88					
C27	EJ1	94					
C28	EJ2	46					
C29	EJ3	49					
C30	EJ4	98					
C31	EK1	95					
C32	EK2	47					
C33	EK3	50					
C34	EK4	99	5.5	0	100	uA	1-3
C35	PGM-N	89	5.5	-113	113	uA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1) (+5 V)	5.5 V 5.5 V	± 55 mV ± 55 mV	C1-C35
Power supply current limit, (Vcc1) (+5 V)	150 mA 148 mA	NA NA	C1-C35
Forcing voltage, (Vin)	5.5 V	± 55 mV	C1-C35
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	C1-C35 C1-C35
Input current measurement	NA NA NA	± 2.0 uA ± 5.0 uA ± 11.3 uA	C1-C17 C18-C34 C35

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IV and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

(e) Apply pattern voltage (VL) to all input pins called out in the test table, except the pin that is connected to vin and leave pin 89 unconnected.

3. Abbreviations used are defined in 6.1.

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TABLE V. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	BO0	82	0.0	-375	-100	mA	1-4
D2	BO1	32					
D3	BO2	33					
D4	BO3	87	0.0	-375	-100	mA	1-4
D5	CSHI	44	0.0	-150	-60	mA	1-4
D6	CSLO	91					
D7	RHI	92					
D8	RLO	93					
D9	EL1	96					
D10	EL2	45					
D11	EL3	48					
D12	EL4	97	0.0	-150	-60	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1)	5.5 V	± 55 mV	D1-D12
(+5 V)	5.5 V	± 55 mV	
Power supply current limit (Vcc1)	150 mA	NA	D1-D12
(+5 V)	148 mA	NA	

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vout)	0.0 V	± 20 mV	D1-D12
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	D1-D12
low, (VL)	0.0 V	± 100 mV	D1-D12
Output current measurement	NA	± 13.7 mA	D1-D4
	NA	± 4.5 mA	D5-D12

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

(e) Apply pattern voltage (VH) to pins 63, 52, 69 and 2. Apply pattern voltage (VL) to all remaining input pins. Leave pin 89 unconnected.

(f) I/O pins are not considered input pins.

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NOTES: - Continued.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VI. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
E	Vcc	7.0	1,51	10,30, 39,40, 90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1) (+5 V)	6.25 V 7.0 V	+0 mV, -63 mV +0 mV, -70 mV	E
Power supply current limit (Vcc1) (+5 V)	150 mA 148 mA	NA NA	E

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table VI.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The following input pins shall be connected to ground: 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 14, 15, 46, 47, 49, 50, 52, 53, 54, 55, 58, 63, 65, 69, 70, 83, 84, 85, 86, 88, 94, 95, 98 and 99.

6. Abbreviations used are defined in 6.1.

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TABLE VII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	A15	55	-18.0	-1.2	-0.2	V	1-3
F2	A14	58					
F3	A13	5					
F4	A12	12					
F5	A11	3					
F6	A10	11					
F7	A09	15					
F8	A08	65					
F9	A07	14					
F10	A06	8					
F11	A05	6					
F12	A04	54					
F13	A03	53					
F14	A02	4					
F15	A01	7					
F16	A00	9					
F17	OE-N/Vpp	70					
F18	UDS-N	69					
F19	LDS-N	63					
F20	CS-N	2					
F21	RAS-N	52	-18.0	-1.2	-0.2	V	1-3

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TABLE VII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
F22	BI0	83	-18.0	-1.2	-0.2	V	1-3
F23	BI1	84					
F24	BI2	85					
F25	BI3	86					
F26	BOE-N	88					
F27	EJ1	94					
F28	EJ2	46					
F29	EJ3	49					
F30	EJ4	98					
F31	EK1	95					
F32	EK2	47					
F33	EK3	50					
F34	EK4	99	-18.0	-1.2	-0.2	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1)	4.5 V	± 45 mV	F1-F34
(+5 V)	4.5 V	± 45 mV	

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Power supply current limit (Vcc1) (+5 V)	150 mA 148 mA	NA NA	F1-F34
Forcing current, (Iin)	-18 mA	± 1.8 mA	F1-F34
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	F1-F34 F1-F34
Input voltage measurement	NA	± 50 mV	F1-F34

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
G1	A15	55	0.5	-20	20	uA	1-4
G2	A14	58					
G3	A13	5					
G4	A12	12					
G5	A11	3					
G6	A10	11					
G7	A09	15					
G8	A08	65					
G9	A07	14					
G10	A06	8					
G11	A05	6					
G12	A04	54					
G13	A03	53					
G14	A02	4					
G15	A01	7					
G16	A00	9					
G17	OE-N/Vpp	70	0.5	-20	20	uA	1-4
G18	UDS-N	69	0.5	-1000	100	uA	1-4
G19	LDS-N	63					
G20	CS-N	2	0.5	-1000	100	uA	1-4

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
G21	RAS-N	52	0.5	-1000	100	uA	1-4
G22	BI0	83					
G23	BI1	84					
G24	BI2	85					
G25	BI3	86					
G26	BOE-N	88	0.5	-1000	100	uA	1-4
G27	EJ1	94	0.5	-600	60	uA	1-4
G28	EJ2	46					
G29	EJ3	49					
G30	EJ4	98					
G31	EK1	95					
G32	EK2	47					
G33	EK3	50					
G34	EK4	99	0.5	-600	60	uA	1-4
G35	PGM-N	89	0.5	-5500	-4500	uA	1-4
G36	D00	35	0.5	-10	10	uA	1-4
G37	D01	37					
G38	D02	41					
G39	D03	43					
G40	D04	42	0.5	-10	10	uA	1-4

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	D05	38	0.5	-10	10	uA	1-4
G42	D06	36					
G43	D07	34					
G44	D08	19					
G45	D09	21					
G46	D10	23					
G47	D11	25					
G48	D12	24					
G49	D13	22					
G50	D14	20					
G51	D15	18	0.5	-10	10	uA	1-4

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1)	5.5 V	± 55 mV	G1-G51
(+5 V)	5.5 V	± 55 mV	
Power supply current limit (Vcc1)	150 mA	NA	G1-G51
(+5 V)	148 mA	NA	

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vil)	0.5 V	± 20 mV	G1-G51
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 G1-G51 G1-G51
Input current measurement	NA NA NA NA NA	± 2 uA ± 55 uA ± 33 uA ± 10 uA ± 1 uA	G1-G17 G18-G26 G27-G34 G35 G36-G51

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

(e) For test G1-G35 all input pins shall be connected to pattern voltage VH, except the pin that is connected to VIL and pins 2, 52, 63, and 69 are connected to pattern voltage (VL). I/O pins are not considered input pin. Leave pin 89 unconnected.

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NOTES: - Continued.

(f) For test G36-G51 connect all input pins to pattern voltage VH, except for pins 89, 52, 63 and 69 which should be connected to pattern voltage VL.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	A15	55	2.7	-20	20	uA	1-4
H2	A14	58					
H3	A13	5					
H4	A12	12					
H5	A11	3					
H6	A10	11					
H7	A09	15					
H8	A08	65					
H9	A07	14					
H10	A06	8					
H11	A05	6					
H12	A04	54					
H13	A03	53					
H14	A02	4					
H15	A01	7					
H16	A00	9					
H17	OE-N/Vpp	70	2.7	-20	20	uA	1-4
H18	UDS-N	69	2.7	-2	20	uA	1-4
H19	LDS-N	63					
H20	CS-N	2	2.7	-2	20	uA	1-4

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TABLE IX. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	RAS-N	52	2.7	-2	20	uA	1-4
H22	BI0	83					
H23	BI1	84					
H24	BI2	85					
H25	BI3	86					
H26	BOE-N	88					
H27	EJ1	94					
H28	EK1	95					
H29	EJ2	46					
H30	EK2	47					
H31	EJ3	49					
H32	EK3	50					
H33	EJ4	98					
H34	EK4	99	2.7	-2	20	uA	1-4
H35	PGM-N	89	2.7	-3080	-2520	uA	1-4
H36	D00	35	2.7	-10	10	uA	1-4
H37	D01	37					
H38	D02	41					
H39	D03	43					
H40	D04	42	2.7	-10	10	uA	1-4

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TABLE IX. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	D05	38	2.7	-10	10	uA	1-4
H42	D06	36					
H43	D07	34					
H44	D08	19					
H45	D09	21					
H46	D10	23					
H47	D11	25					
H48	D12	24					
H49	D13	22					
H50	D14	20					
H51	D15	18	2.7	-10	10	uA	1-4

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1) (+5 V)	5.5 V 5.5 V	± 55 mV ± 55 mV	H1-H51
Power supply current limit (Vcc1) (+5 V)	150 mA 148 mA	NA NA	H1-H51
Forcing voltage, (Vih)	2.7 V	± 120 mV	H1-H51

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H51 H1-H51
Input current measurement	NA NA NA NA	± 2.0 uA ± 1.1 uA ± 0.176 mA ± 1.0 uA	H1-H17 H18-H34 H35 H36-H51

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

(e) All input pins shall be connected to VL, except the pin that is connected to Vih. I/O are not considered input pin.

(f) For test conditions H36-H51, connect pins 2 and 70 to patern voltage (VH).

(g) For test conditions H1-H35 leave pin 89 unconnected.

3. Abbreviations used are defined in 6.1.

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TABLE X. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	D00	35	2.7	-10	10	uA	1-3
J2	D01	37					
J3	D02	41					
J4	D03	43					
J5	D04	42					
J6	D05	38					
J7	D06	36					
J8	D07	34					
J9	D08	19					
J10	D09	21					
J11	D10	23					
J12	D11	25					
J13	D12	24					
J14	D13	22					
J15	D14	20					
J16	D15	18	2.7	-10	10	uA	1-3
J17	BO0	82	2.7	-5	50	uA	1-3
J18	BO1	32					
J19	BO2	33					
J20	BO3	87	2.7	-5	50	uA	1-3

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NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1) (+5 V)	5.5 V 5.5 V	± 55 mV ± 55 mV	J1-J20
Power supply current limit (Vcc1) (+5 V)	150 mA 148 mA	NA NA	J1-J20
Forcing voltage, (Voh)	2.7 V	± 135 mV	J1-J20
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J20 J1-J20
Output current measurement	NA NA	± 1.0 uA ± 2.75 uA	J1-J16 J17-J20

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table X and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

(e) Apply pattern voltage (VH) to pins 52 and 88. Apply pattern voltage (VL) to remaining input pins.

3. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
K1	D00	35	0.4	-10	10	uA	1-3
K2	D01	37					
K3	D02	41					
K4	D03	43					
K5	D04	42					
K6	D05	38					
K7	D06	36					
K8	D07	34					
K9	D08	19					
K10	D09	21					
K11	D10	23					
K12	D11	25					
K13	D12	24					
K14	D13	22					
K15	D14	20					
K16	D15	18	0.4	-10	10	uA	1-3
K17	BO0	82	0.55	-50	5	uA	1-3
K18	BO1	32					
K19	BO2	33					
K20	BO3	87	0.55	-50	5	uA	1-3

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NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1) (+5 V)	5.5 V 5.5 V	± 55 mV ± 55 mV	K1-K20
Power supply current limit (Vcc1) (+5 V)	150 mA 148 mA	NA NA	K1-K20
Forcing voltage, (Vol)	0.4 V 0.55 V	± 20 mV ± 28 mV	K1-K16 K17-K20
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	K1-K20 K1-K20
Output current measurement	NA NA	± 1.0 uA ± 2.8 uA	K1-K16 K17-K20

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

(e) Apply pattern voltage (VL) to all input pins, except 52 and 88 to (VH) and pin connected to VOL.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	D00	35	-24.0	3.7	4.5	V	1-4
L2	D01	37					
L3	D02	41					
L4	D03	43					
L5	D04	42					
L6	D05	38					
L7	D06	36					
L8	D07	34					
L9	D08	19					
L10	D09	21					
L11	D10	23					
L12	D11	25					
L13	D12	24					
L14	D13	22					
L15	D14	20					
L16	D15	18	-24.0	3.7	4.5	V	1-4
L17	BO0	82	-3.0	2.4	4.5	V	1-4
L18	BO1	32					
L19	BO2	33					
L20	BO3	87	-3.0	2.4	4.5	V	1-4

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TABLE XII. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	CSHI	44	-1.0	2.5	4.5	V	1-4
L22	CSLO	91					
L23	RHI	92					
L24	RLO	93					
L25	EL1	96					
L26	EL2	45					
L27	EL3	48					
L28	EL4	97	-1.0	2.5	4.5	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1)	4.5 V	± 45 mV	L1-L28
(+5 V)	4.5 V	± 45 mV	
Power supply current limit (Vcc1)	150 mA	NA	L1-L28
(+5 V)	148 mA	NA	
Forcing current, (Ioh)	-24 mA	± 240 uA	L1-L16
	-3 mA	± 30 uA	L17-L20
	-1 mA	± 10 uA	L21-L28

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 2.0 V 0.8 V	 ± 20 mV ± 8 mV	 L1-L28 L1-L28
Output voltage measurement	NA NA NA	± 40 mV ± 105 mV ± 100 mV	L1-L16 L17-L20 L21-L28

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

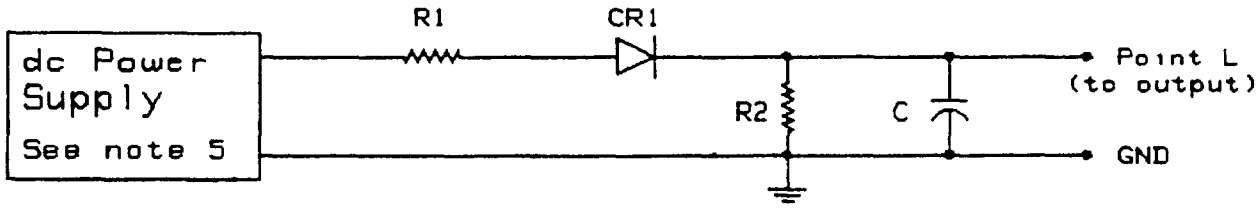
(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

(e) For test conditions L1-L16 apply pattern voltage (VH) to pins 53, 4, 9, and 89. Apply pattern voltage (VL) to remaining input pins. For test conditions L17-L28 apply pattern voltage (VH) to pins 52, 2, 63 and 69. Apply pattern voltage (VL) to remaining input pins.

3. The load shall be as indicated on figure 7.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 300 ± 30 pF (including probe and parasitic capacitance) for sample dynamic, propagation delay time, margin and disable time tests.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk test.

(c) C is not required for any other tests.

3. Select R1 for a current flow of Iol when point L is held at Vol with the load power supply set at 2.5 V ± 1 percent. The values of Iol and Vol for each output are listed below.

4. The value of R2 for each output load shall be as specified below.

5. The load power supply voltage shall be 2.1V ± 1 percent for enable/disable time test and 2.5V ± 1 percent for all other test.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
18,19,20,21,22,23, 24,25,34,35,36,37, 38,41,42,43	154	24	0.5
32,33,82,87	800	48	0.55
44,45,48,91,92,93, 96,97	2500	20	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

FIGURE 7. Logic load circuit.

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TABLE XIII. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	D00	35	24.0	0.0	0.5	V	1-4
M2	D01	37					
M3	D02	41					
M4	D03	43					
M5	D04	42					
M6	D05	38					
M7	D06	36					
M8	D07	34					
M9	D08	19					
M10	D09	21					
M11	D10	23					
M12	D11	25					
M13	D12	24					
M14	D13	22					
M15	D14	20					
M16	D15	18	24.0	0.0	0.5	V	1-4
M17	BO0	82	48.0	0.0	0.55	V	1-4
M18	BO1	32					
M19	BO2	33					
M20	BO3	87	48.0	0.0	0.55	V	1-4

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Check the source to verify that this is the current version before use.

TABLE XIII. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M21	CSHI	44	20.0	0.0	0.5	V	1-4
M22	CSLO	91					
M23	RHI	92					
M24	RLO	93					
M25	EL1	96					
M26	EL2	45					
M27	EL3	48					
M28	EL4	97	20.0	0.0	0.5	V	1-4

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1)	4.5 V	± 45 mV	M1-M28
(+5 V)	4.5 V	± 45 mV	
Power supply current limit (Vcc1)	150 mA	NA	M1-M28
(+5 V)	148 mA	NA	
Forcing current, (Iol)	24 mA	± 240 uA	M1-M16
	48 mA	± 480 uA	M17-M20
	20 mA	± 200 uA	M21-M28

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 2.0 V 0.8 V	 ± 20 mV ± 8 mV	 M1-M28 M1-M28
Output voltage measurement	NA NA NA	± 25.0 mV ± 27.5 mV ± 25.0 mV	M1-M16 M17-M20 M21-M28

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XIII and the voltage then measured.

(e) Apply pattern voltage (VL) to pins 2, 3, 5, 6, 7, 8, 9, 11, 12, 14, 15, 52, 54, 55, 58, 63, 65, 69 and 88. Apply pattern voltage (VH) to pins 83, 84, 85, 86, 89, 53, 4, 94, 95, 46, 47, 49, 50, 98, and 99.

3. The load shall be as indicated on figure 7.

4. Abbreviations used are defined in 6.1.

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TABLE XIV. Power supply current test.

Test cond	Parameter	A to pin	B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
N1	Icc (logic quiescent)	51	1	0	96	mA	1-4
N2	Idd (EPROM quiescent)	1	51	0	100	mA	1-4
N3	Icc (logic disable)	51	1	0	96	mA	1-4
N4	Idd (EPROM disable)	1	51	0	200	uA	1-4

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1) (+5 V)	5.5 V 5.5 V	± 55 mV ± 55 mV	N1-N4
Power supply current limit (Vcc1) (+5 V)	150 mA 148 mA	NA NA	N1-N4
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	N1-N4 N1-N4
Power supply current measurement	NA NA NA NA	± 4.8 mA ± 5.0 mA ± 4.8 mA ± 0.08 mA	N1 N2 N3 N4

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

(c) In the setup conditions of table XIV, 'A' shall correspond to the positive terminal of the power supply, and 'B' shall correspond to the positive terminal of the dc ammeter.

3. Quiescent power supply current is measured with UDS-N, CS-N, LDS-N and RAS-N tied to logic low. Disable power supply current is measured with CS-N and RAS-N tied to logic high.

4. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P	Functional test minimum power supply voltage maximum pattern rate	1-6

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1) (+5 V)	4.5 V 4.5 V	± 45 mV ± 45 mV	P
Power supply current limit (Vcc1) (+5 V)	150 mA 148 mA	NA NA	P
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	P P
Output voltage verification high, (Voh) low, (Vol)	NA NA	± 105 mV ± 25 mV	P P

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in note 3. The output compare levels shall be as follows:

(for output signals D15-D0)

High level: 2.0 V ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ 0.8 V

(for output signals BO3-BO0)

High level: 2.0 V ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ 0.8 V

(for output signals EL4-EL1)

High level: 2.0 V ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ 0.8 V

3. The memory test pattern for UV EPROMs is specified below:

Address	Data
0000-000F	00 FF 00 01 02 04 08 10 20 40 80 00 00 FF FF FE
0010-001F	FD FB F7 EF DF BF 7F 55 AA 55 AA 55 AA 55 AA 55
0020-FFFF	AA 55 ...

4. This test shall operate the module at no more than 1 MHz clock rate, and only a portion of the memory of each UV EPROM is required for verification. The dynamic test shall test 100 percent of the module's memory.

5. The information below is the proper logic states for the output signals BO3-BO0 and EL4-EL1:

Input		Output
BOE-N	BI3-BI0	BO3-BO0
L	L	H
L	H	L
H	X	Z

Input		Output
EJ4-EJ1	EK4-EK1	EL4-EL1
L	L	H
H	L	H
L	H	H
H	H	L

6. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate minimum power supply	1-10
Q2	Dynamic test, sample maximum pattern rate maximum power supply	1-10

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1) (+5 V) (Vcc1) (+5 V)	4.5 V 4.5 V 5.5 V 5.5 V	± 45 mV ± 45 mV ± 55 mV ± 55 mV	Q1 Q2
Power supply current limit (Vcc1) (+5 V)	150 mA 742 mA	NA NA	Q1,Q2
Input voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 120 mV ± 20 mV	 Q1,Q2 Q1,Q2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 Q1 Q2 Q1,Q2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in note 3. The output compare levels shall be as follows:

(for output signals D15-D00)

High level: 2.0 V $\leq$ Voh $\leq$ Vcc
Low level: 0.0 V $\leq$ Vol $\leq$ 0.8 V

(for output signals BO3-BO0)

High level: 2.0 V $\leq$ Voh $\leq$ Vcc
Low level: 0.0 V $\leq$ Vol $\leq$ 0.8 V

(for output signals EL1-EL4)

High level: 2.0 V $\leq$ Voh $\leq$ Vcc
Low level: 0.0 V $\leq$ Vol $\leq$ 0.8 V

3. The memory test pattern for the UV EPROMs is specified below:

Address	Data
0000-000F	00 FF 00 01 02 04 08 10 20 40 80 00 00 FF FF FE
0010-001F	FD FB F7 EF DF BF 7F 55 AA 55 AA 55 AA 55 AA 55
0020-FFFF	AA 55 ...

4. This dynamic test shall test 100 percent of the module's memory. The maximum operating frequency is 5 MHz.

5. The information below is the proper logic states for the output signals BO3-BO0 and EL4-EL1:

Input		Output
BOE-N	BI3-BI0	BO3-BO0
L	L	H
L	H	L
H	X	Z

Input		Output
EJ4-EJ1	EK4-EK1	EL4-EL1
L	L	H
H	L	H
L	H	H
H	H	L

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NOTES: - Continued.

6. The load circuit for this test is specified on figure 7.

7. This test shall be performed as indicated in table XVI. The dynamic parameters are found in table XVII.

8. Input/Output (I/O) waveforms are as referenced on figure 8.

9. OE-N/Vpp may be delayed up to tCE-tOE after the falling edge of LDS-N, UDS-N, CS-N and RAS-N without impact on tCE.

10. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref	Parameter	Minimum limit			Maximum limit			Units	Fig
		-55°C	25°C	85°C	-55°C	25°C	85°C		
1	A15-A00 to D15-D00, tACC	-	-	-	180	180	180	nS	8
2	UDS-N, LDS-N to D15-D00, tCE	-	-	-	180	180	180	nS	8
2	RAS-N, CS-N to D15-D00, tCE	-	-	-	180	180	180	nS	8
3	OE-N/Vpp to D15-D00, tOE	-	-	-	82	82	82	nS	8

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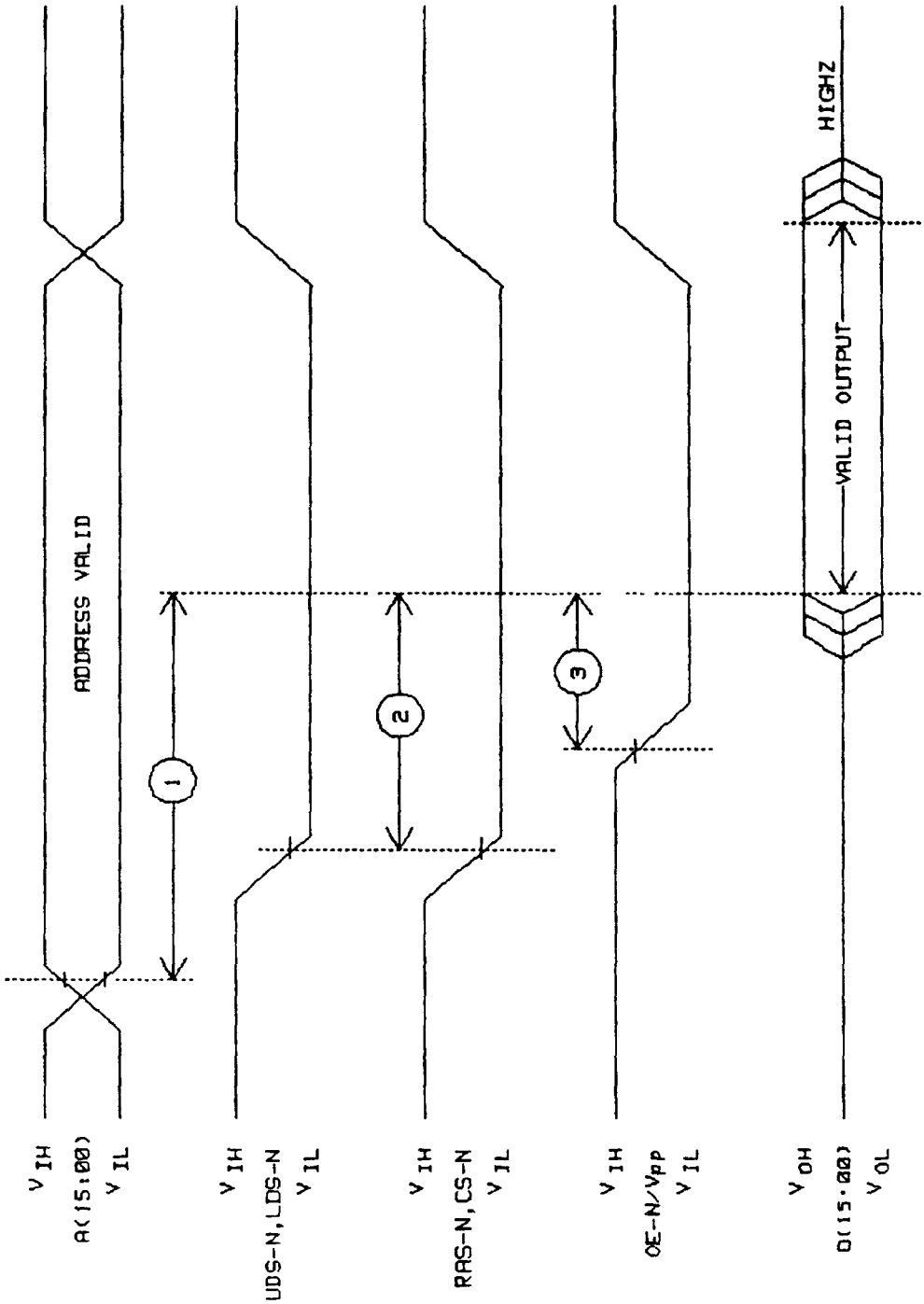


FIGURE 8. Dynamic waveforms for table XVII.

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TABLE XVIII. Enable/Disable time test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Fig
High impedance to high (tPZH)							
R1	D00	35	0	40	nS	1-4	9
R2	D01	37					
R3	D02	41					
R4	D03	43					
R5	D04	42					
R6	D05	38					
R7	D06	36					
R8	D07	34					
R9	D08	19					
R10	D09	21					
R11	D10	23					
R12	D11	25					
R13	D12	24					
R14	D13	22					
R15	D14	20					
R16	D15	18					
R17	BO0	82					
R18	BO1	32					
R19	BO2	33					
R20	BO3	87	0	40	nS	1-4	9

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TABLE XVIII. Enable/Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Fig
High impedance to low (tPZL)							
R21	D00	35	0	40	nS	1-4	9
R22	D01	37					
R23	D02	41					
R24	D03	43					
R25	D04	42					
R26	D05	38					
R27	D06	36					
R28	D07	34					
R29	D08	19					
R30	D09	21					
R31	D10	23					
R32	D11	25					
R33	D12	24					
R34	D13	22					
R35	D14	20					
R36	D15	18					
R37	BO0	82					
R38	BO1	32					
R39	BO2	33					
R40	BO3	87	0	40	nS	1-4	9

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TABLE XVIII. Enable/Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
High to high impedance (tPHZ)							
R41	D00	35	0	40	nS	1-4	9
R42	D01	37					
R43	D02	41					
R44	D03	43					
R45	D04	42					
R46	D05	38					
R47	D06	36					
R48	D07	34					
R49	D08	19					
R50	D09	21					
R51	D10	23					
R52	D11	25					
R53	D12	24					
R54	D13	22					
R55	D14	20					
R56	D15	18					
R57	BO0	82					
R58	BO1	32					
R59	BO2	33					
R60	BO3	87	0	40	nS	1-4	9

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TABLE XVIII. Enable/Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Fig
Low to high impedance (tPLZ)							
R61	D00	35	0	40	nS	1-4	9
R62	D01	37					
R63	D02	41					
R64	D03	43					
R65	D04	42					
R66	D05	38					
R67	D06	36					
R68	D07	34					
R69	D08	19					
R70	D09	21					
R71	D10	23					
R72	D11	25					
R73	D12	24					
R74	D13	22					
R75	D14	20					
R76	D15	18					
R77	BO0	82					
R78	BO1	32					
R79	BO2	33					
R80	BO3	87	0	40	nS	1-4	9

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NOTES:

1. Enable/Disable time test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1) (+5 V)	5.0 V 5.0 V	± 50 mV ± 50 mV	R1-R80
Power supply current limit (Vcc1) (+5 V)	150 mA 742 mA	NA NA	R1-R80
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	R1-R80 R1-R80
Enable/Disable time measurements	NA	± 2 nS	R1-R80

2. The following procedure is required to perform this test:

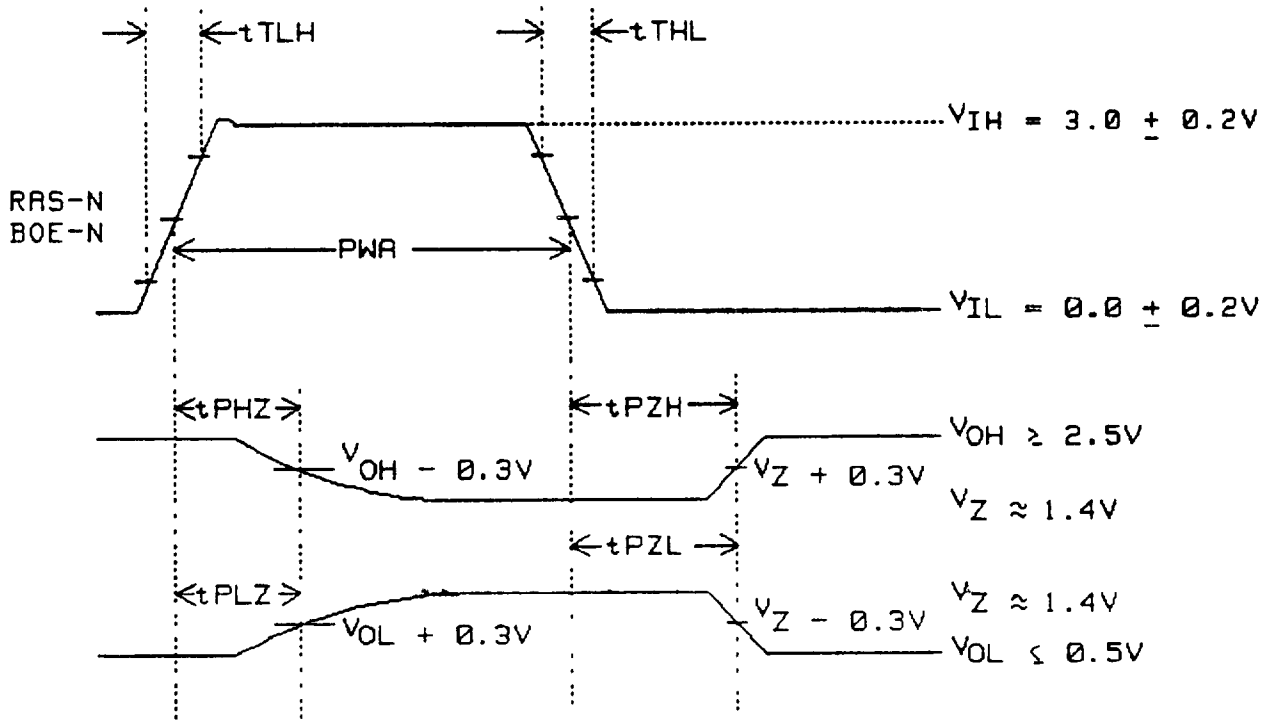
(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table. The module inputs shall meet the minimum input timing requirements as shown on figure 9.

(b) For test conditions R1-R20 and R41-R60 apply pattern voltage (VL) to all input pins, except pins 9 and 89 to (VH). For test conditions R21-R40 and R61-R80 apply pattern voltage (VL) to all input pins, except pin 89 to (VH).

3. The load shall be as indicated on figure 7.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. RAS-N and BOE-N pulse parameters:
PWA = 500 nS, tTHL ≤ 10 nS, tTLH ≤ 10 nS.

FIGURE 9. Enable/Disable time test waveforms for table XVIII.

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TABLE XIX. Capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Input capacitance (Cin)						
S1	A15	55	2	30	pF	1-3
S2	A14	58				
S3	A13	5				
S4	A12	12				
S5	A11	3				
S6	A10	11				
S7	A09	15				
S8	A08	65				
S9	A07	14				
S10	A06	8				
S11	A05	6				
S12	A04	54				
S13	A03	53				
S14	A02	4				
S15	A01	7				
S16	A00	9				
S17	OE-N/Vpp	70				
S18	BOE-N	88				
S19	PGM-N	89	2	30	pF	1-3
S20	CS-N	2	2	15	pF	1-3

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TABLE XIX. Capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
S21	RAS-N	52	2	15	pF	1-3
S22	BI0	83				
S23	BI1	84				
S24	BI2	85				
S25	BI3	86				
S26	UDS-N	69				
S27	EJ1	94				
S28	EJ2	46				
S29	EJ3	49				
S30	EJ4	98				
S31	EK1	95				
S32	EK2	47				
S33	EK3	50				
S34	EK4	99				
S35	LDS-N	63	2	15	pF	1-3
Output capacitance (Cout)						
S36	D00	35	2	20	pF	1-3
S37	D01	37				
S38	D02	41				
S39	D03	43				
S40	D04	42	2	20	pF	1-3

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TABLE XIX. Capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
S41	D05	38	2	20	pF	1-3
S42	D06	36				
S43	D07	34				
S44	D08	19				
S45	D09	21				
S46	D10	23				
S47	D11	25				
S48	D12	24				
S49	D13	22				
S50	D14	20				
S51	D15	18	2	20	pF	1-3
S52	CSHI	44	2	35	pF	1-3
S53	CSLO	91				
S54	RHI	92				
S55	RLO	93				
S56	BO0	82				
S57	BO1	32				
S58	BO2	33				
S59	BO3	87				
S60	EL1	96	2	35	pF	1-3

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TABLE XIX. Capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
S61	EL2	45	2	35	pF	1-3
S62	EL3	48	|	|	|	|
S63	EL4	97	2	35	pF	1-3

NOTES:

1. Capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	S1-S63
Capacitance meter voltage	15 mV	± 1.5 mV	S1-S63
Capacitance measurement	NA NA NA NA	± 1.4 pF ± 0.65 pF ± 0.9 pF ± 1.65 pF	S1-S19 S20-S35 S36-S51 S52-S63

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIX.

3. Abbreviations used are defined in 6.1.

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TABLE XX. Output crosstalk test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
Crosstalk high level							
T1	D00	35	0	550	mV	1-4	10,11
T2	D01	37					
T3	D02	41					
T4	D03	43					
T5	D04	42					
T6	D05	38					
T7	D06	36					
T8	D07	34					
T9	D08	19					
T10	D09	21					
T11	D10	23					
T12	D11	25					
T13	D12	24					
T14	D13	22					
T15	D14	20					
T16	D15	18	0	550	mV	1-4	10,11
T17	BO0	82	0	400	mV	1-4	10,11
T18	BO1	32					
T19	BO2	33					
T20	BO3	87	0	400	mV	1-4	10,11

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TABLE XX. Output crosstalk test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
T21	CSHI	44	0	400	mV	1-4	10,11
T22	CSLO	91					
T23	RHI	92					
T24	RLO	93					
T25	EL1	96					
T26	EL2	45					
T27	EL3	48					
T28	EL4	97	0	400	mV	1-4	10,11
Crosstalk low level							
T29	D00	35	0	550	mV	1-4	10,11
T30	D01	37					
T31	D02	41					
T32	D03	43					
T33	D04	42					
T34	D05	38					
T35	D06	36					
T36	D07	34					
T37	D08	19					
T38	D09	21					
T39	D10	23					
T40	D11	25	0	550	mV	1-4	10,11

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TABLE XX. Output crosstalk test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
T41	D12	24	0	550	mV	1-4	10,11
T42	D13	22					
T43	D14	20					
T44	D15	18	0	550	mV	1-4	10,11
T45	BO0	82	0	400	mV	1-4	10,11
T46	BO1	32					
T47	BO2	33					
T48	BO3	87					
T49	CSHI	44					
T50	CSLO	91					
T51	RHI	92					
T52	RLO	93					
T53	EL1	96					
T54	EL2	45					
T55	EL3	48					
T56	EL4	97	0	400	mV	1-4	10,11

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NOTES:

1. Output crosstalk, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1) (+5 V)	5.0 V 5.0 V	± 50 mV ± 50 mv	T1-T56
Power supply current limit (Vcc1) (+5 V)	150 mA 742 mA	NA NA	T1-T56
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 200 mV ± 100 mV	T1-T56 T1-T56
Output crosstalk measurement	NA	± 20 mV	T1-T56

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XX, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 11.

(e) The maximum probe capacitance is 10 pF.

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NOTES - Continued.

3. The test circuit shall be as indicated on figure 10. All outputs shall be loaded as specified on figure 7.

4. Abbreviations used are defined in 6.1.

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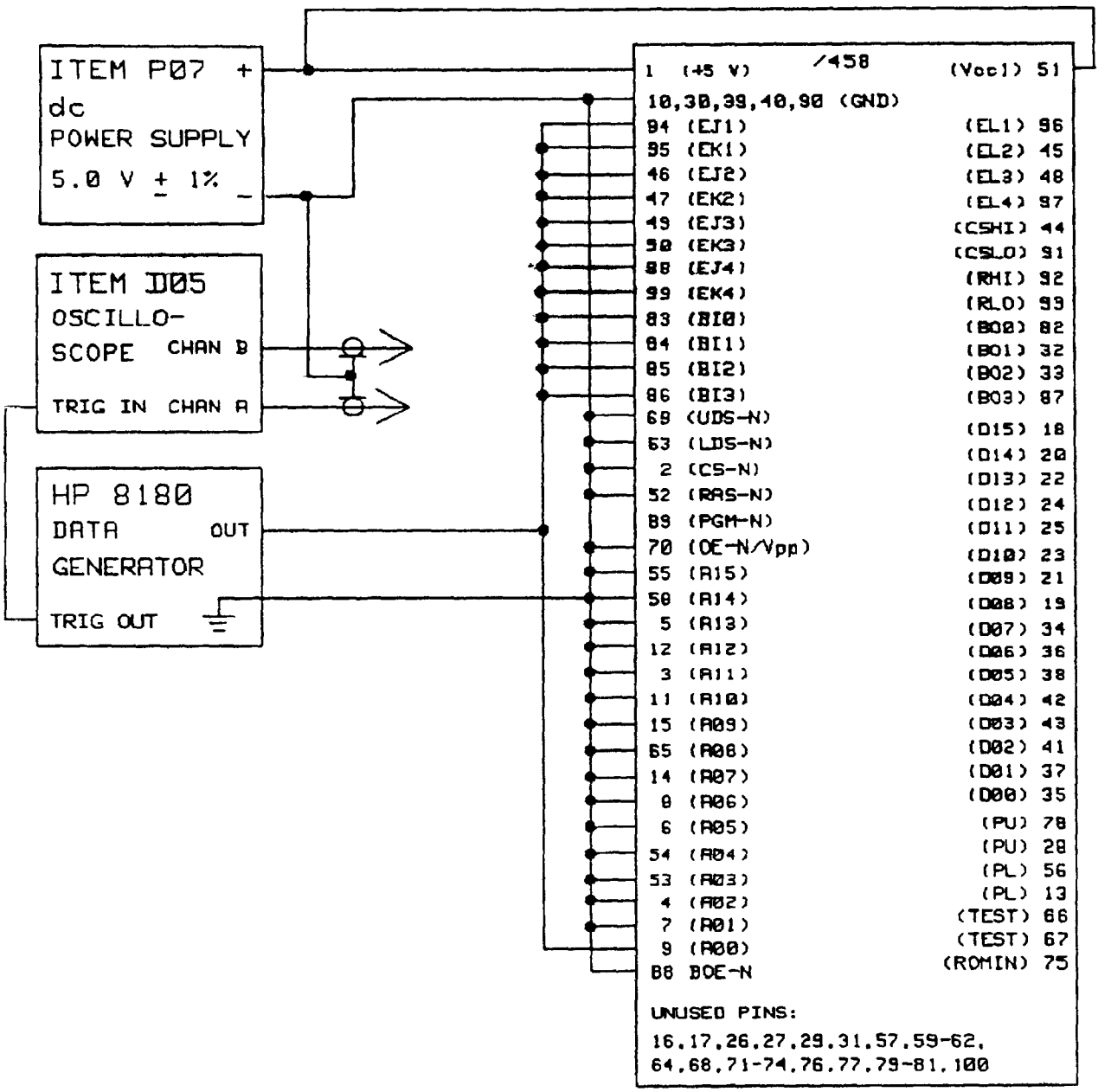


FIGURE 10. Output crosstalk, high and low level, test circuit for table XX.

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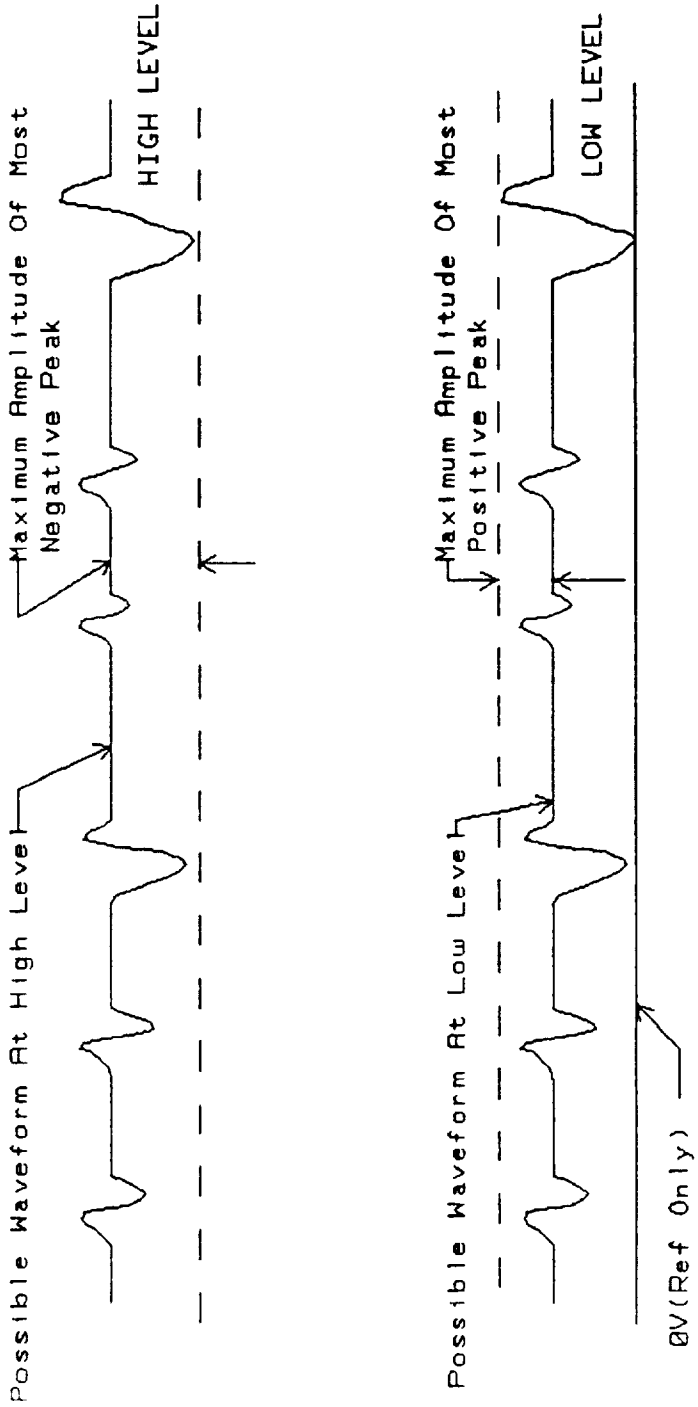


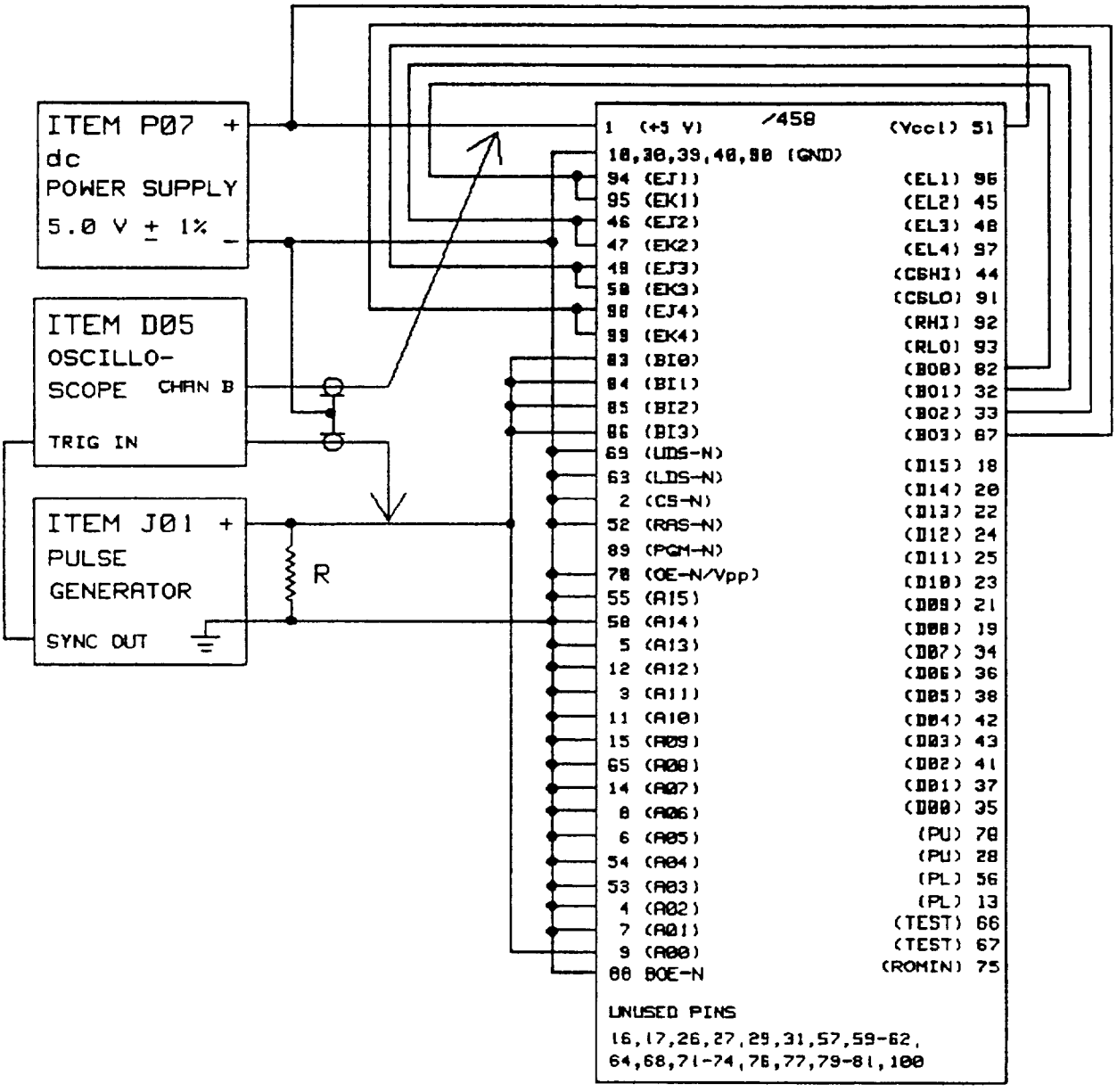
FIGURE 11. Output crosstalk, high and low level, test waveforms for table XX.

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TABLE XXI. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits		Units	Notes
			Min	Max		
U	PSTA	1,51	0	400	mV	1-4

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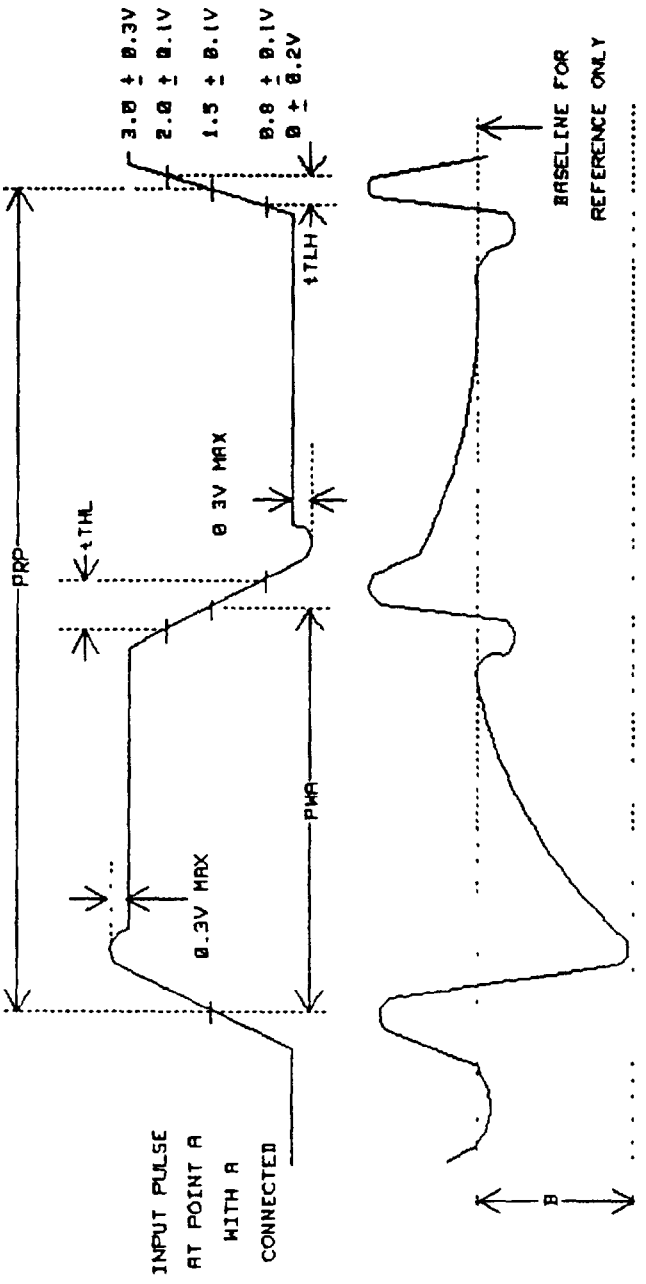


NOTE:

1. R = 51 Ohms ± 5 percent, 1 W.

FIGURE 12. Power supply transient amplitude test circuit for table XXI.

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NOTE:

1. Pulse parameters: PRP = 1000 nS, PWA = 500 nS.
   tTHL ≤ 10 nS, tTLH ≤ 10 nS.

FIGURE 13. Power supply transient amplitude waveform for table XXI.

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TABLE XXII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
V1	PSFC	1	3.71	6.89	uF	1-3
V2	PSFC	51	3.43	6.37	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	V1,V2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	V1,V2
Capacitance measurement	NA NA	± 159 nF ± 147 nF	V1 V2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIII. Propagation delay time test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
High to low (tPHL)							
W1	EJ1 to EL1	96	0	34	nS	1-5	14
W2	EJ2 to EL2	45					
W3	EJ3 to EL3	48					
W4	EJ4 to EL4	97					
W5	EK1 to EL1	96					
W6	EK2 to EL2	45					
W7	EK3 to EL3	48					
W8	EK4 to EL4	97					
W9	BI0 to BO0	82					
W10	BI1 to BO1	32					
W11	BI2 to BO2	33					
W12	BI3 to BO3	87	0	34	nS	1-5	14
Low to high (tPLH)							
W13	EJ1 to EL1	96	0	34	nS	1-5	14
W14	EJ2 to EL2	45					
W15	EJ3 to EL3	48					
W16	EJ4 to EL4	97					
W17	EK1 to EL1	96					
W18	EK2 to EL2	45					
W19	EK3 to EL3	48	0	34	nS	1-5	14

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TABLE XXIII. Propagation delay time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
W20	EK4 to EL4	97	0	34	nS	1-5	14
W21	BI0 to BO0	82					
W22	BI1 to BO1	32					
W23	BI2 to BO2	33					
W24	BI3 to BO3	87	0	34	nS	1-5	14

NOTES:

1. Propagation delay time test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc1) (+5 V)	5.5 V 5.5 V	± 55 mV ± 55 mV	W1-W24
Power supply current limit (Vcc1) (+5 V)	150 mA 742 mA	NA NA	W1-W24
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 120 mV ± 20 mV	W1-W24 W1-W24
Output voltage verification high, (VOH) low, (VOL)	NA NA	± 120 mV ± 20 mV	W1-W24 W1-W24

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. The inputs shall meet the requirements as shown on figure 14.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. All outputs shall be loaded as specified on figure 7.

4. Disable the output transceiver by connecting RAS-N at a logic high.

5. Abbreviations used are defined in 6.1.

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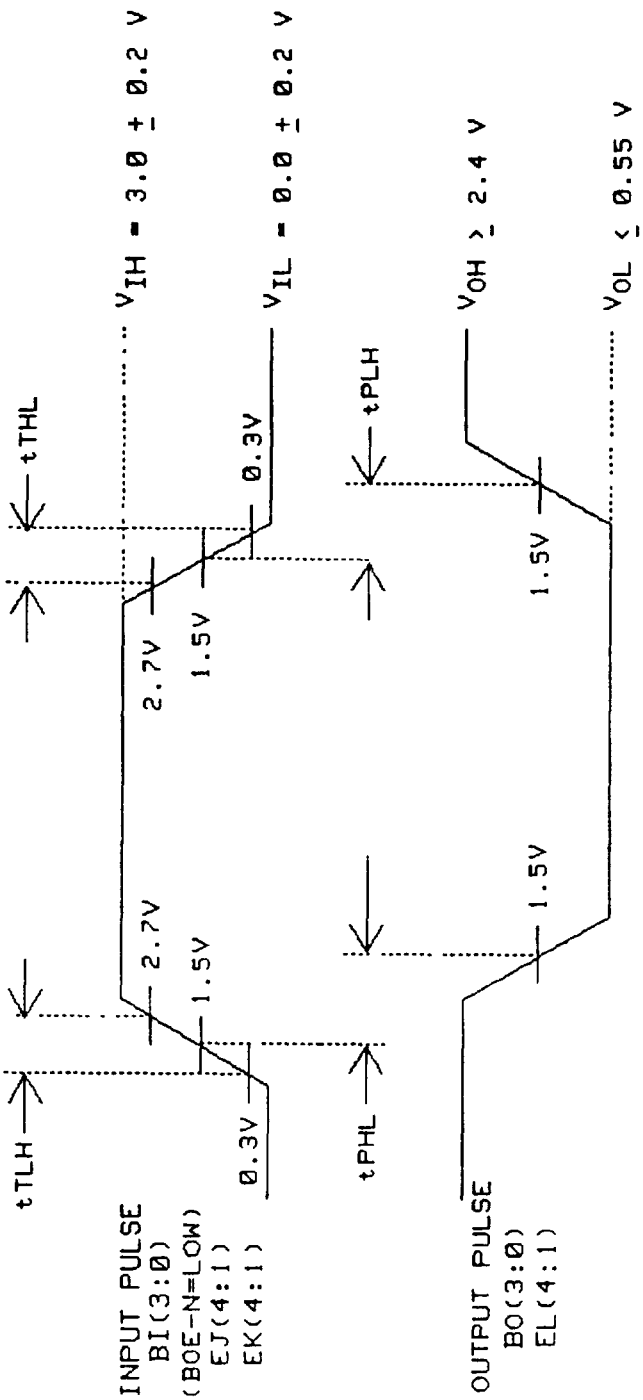


FIGURE 14. Propagation delay test waveforms for table XXIII.

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TABLE XXIV. Erasing test.

Test cond	Parameter	Test description	Notes
X	Erasing	----------see notes----------	1-4

NOTES:

1. The entire memory array is erased after exposure to ultraviolet light at a wavelength of 2537 angstroms. A complete erasure is assured after a minimum of 20 minutes exposure using 12,000 uW/cm ** 2 intensity lamp spaced one inch away from the UV EPROM. Minimum erase time for lamps at other intensity ratings can be calculated from the minimum integrated erasure dose of 15 W-sec/cm ** 2. To prevent unintentional erasure, an opaque label is recommended to cover the window of the programmed UV EPROM.

2. As one of the qualification tests, the two UV EPROMs are to be erased before the programmability test Y and the margin test Z.

3. After erasing the memory of the two UV EPROMs, all address locations of the module shall be tested to read active high. In general, a test fixture is developed to interface the module to a device programmer. In this fashion, the programmer will generally verify the selected device as being erased before programming. Note, however, it is not recommended that the module be erased unless programming is to immediately follow. The erased cycles should be kept at a minimum because the number of erasures that the module has been subjected to determines the deterioration of the module's capacity to hold memory.

4. Abbreviations used are defined in 6.1.

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TABLE XXV. Programming test.

Test cond	Parameter	Test description	Notes
Y	Programming	-------see notes-------	1-3

NOTES:

1. Programming test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (+5 V) (Vcc1) (OE-N/Vpp)	5.0 V 6.0 V 12.5 V	± 55 mV ± 60 mV ± 125 mV	Y
Power supply current limit, (+5 V) (Vcc1) (OE-N/Vpp)	742 mA 150 mA 75 mA	NA NA NA	Y
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	Y Y
Output voltage verification high, (VOH) low, (VOL)	NA NA	± 40 mV ± 25 mV	Y Y

2. This qualification test is to follow test condition X and to preclude test condition Z. The memory test pattern shall be as specified below:

Address	Data
0000-000F	00 FF 00 01 02 04 08 10 20 40 80 00 00 FF FF FE
0010-001F	FD FB F7 EF DF BF 7F 55 AA 55 AA 55 AA 55 AA 55
0020-FFFF	AA 55...

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NOTES: - Continued.

3. Abbreviations used are defined in 6.1.

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TABLE XXVI. Margin test.

Test cond	Parameter	Test description	Notes
Z	Margin	---------see notes---------	1-8

NOTES:

1. Margin test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (+5 V) (Vcc1)	5.5 V 5.8 V	± 55 mV ± 58 mV	Z
Power supply current limit (+5 V) (Vcc1)	742 mA 150 mA	NA NA	Z
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	Z Z
Output voltage verification high, (VOH) low, (VOL)	NA NA	± 40 mV ± 25 mV	Z Z

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) All memory shall be tested to read out the test pattern as indicated in note 3 for all address locations. The output compare levels for output signals D15-D00 shall be as follows:

High level: $2.0\ V \leq Vol \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.8\ V$

3. The memory test pattern for UV EPROMs is specified below:

Address	Data
0000-000F	00 FF 00 01 02 04 08 10 20 40 80 00 00 FF FF FE
0010-001F	FD FB F7 EF DF BF 7F 55 AA 55 AA 55 AA 55 AA 55
0020-FFFF	AA 55 ...

4. The maximum operating frequency is 5 MHz.

5. The load circuit for this test is specified on figure 7.

6. The I/O waveforms are as referenced on figure 15.

7. This qualification test is to preclude test condition Z and to follow test condition Y.

8. Abbreviations used are defined in 6.1.

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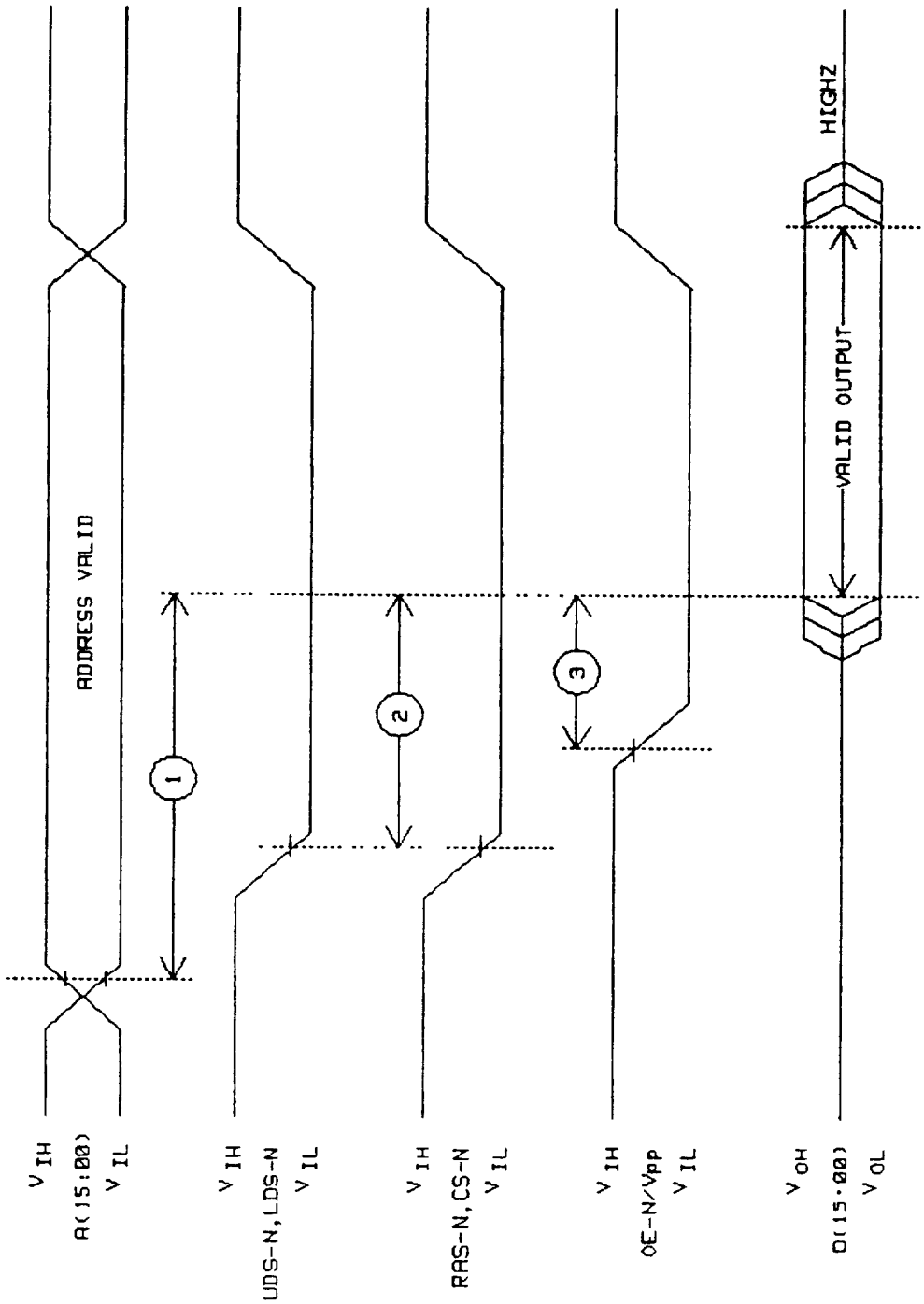


FIGURE 15. Module read waveforms for table XXVI.

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NOTES:

1. tACC = 180 nS maximum.

2. tCE = 180 nS maximum.

3. tOE = 82 nS maximum.

4. OE-N/Vpp may be delayed up to tCE-tOE after the falling edge of LDS-N, UDS-N, CS-N and RAS-N without impact on tCE.

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APPENDIX

MODULE PROGRAMMING

10. SCOPE

10.1 Scope. This appendix contains information regarding the module programming. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

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APPENDIX

30. Module programming. A flow chart description of how to program the module is shown on figures 1a and 1b. This description, along with the timing diagram given on figures 1c and 2, and table I, will provide enough information to accomplish the task of programming the module. There are two UV EPROMs on the module, and each of them contains 64K bytes of memory. Qualification tests X and Y test whether the module can be programmed and erased. Qualification test Z is an additional test which is designated as the margin test. This test verifies the reading performance of all the memory contained on the module following programming with the UV EPROM power pin (Vcc1) set at 5.8 V.

30.1 Fast programming algorithm. In programming the module, each UV EPROM device is programmed separately. When signal UDS-N is logic low, it selects the upper UV EPROM device, and when signal LDS-N is logic low, it selects the lower UV EPROM device. Two CS-N pulse widths are used to program; these are initial and over-program. Address A15-A00 are set to address the desired byte with RAS-N and PGM-N at a logic low. Vcc1 is raised to 6.0 V, and OE-N/Vpp is raised to 12.5 V. Vcc1 must be applied simultaneously or before OE-N/Vpp and removed simultaneously or after OE-N/Vpp. The first CS-N pulse is 1 mS. The programmed byte is then verified. In verification, the transceiver directional control signal PGM-N must be at a logic high. Note, however, this signal PGM-N has an internal pullup resistor on the module. After verification, an over-programming CS-N pulse is applied for 3 mS. If the device fails to program after the first 1 mS pulse, then up to 25 successive 1 mS pulses are applied with a verification after each pulse. When the byte passes the verification step, the over-programming pulse width is 3 times the number of 1 mS pulses required earlier (75 mS max). If the device fails to verify after 25 of 1 mS pulses have been applied, it is considered as failed. After the first byte is progammed, the address bus is set to the next address repeating the algorithm above until all required addresses are programmed. The Vcc1 and OE-N/Vpp are lower to 5.0 V. All bytes are subsequently read to compare with the original data to determine if the device passes or fails. The above procedure is performed twice to accomplish the programming of both UV EPROM devices where the only difference is whether the upper UV EPROM device is selected by UDS-N at logic low and LDS-N at logic high or the lower UV EPROM is selected with LDS-N at logic low and UDS-N at logic high.

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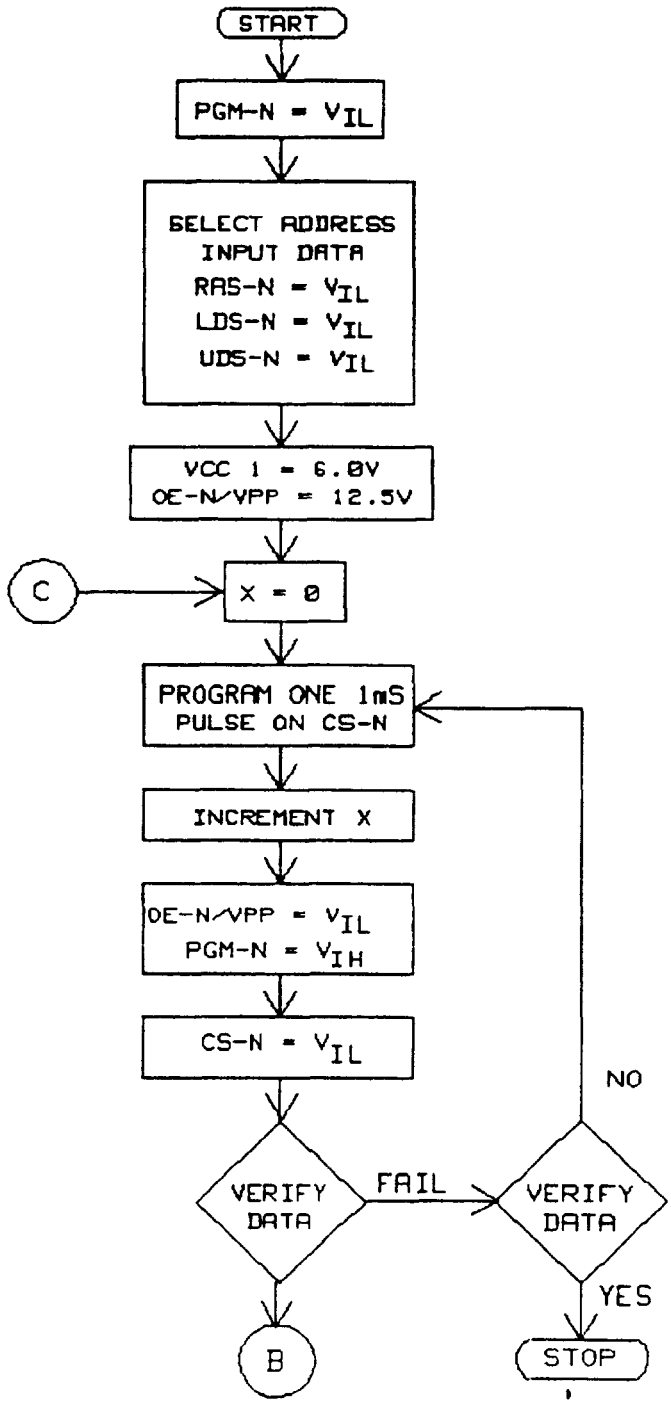


FIGURE 1a. Programming algorithm for UV EPROM.

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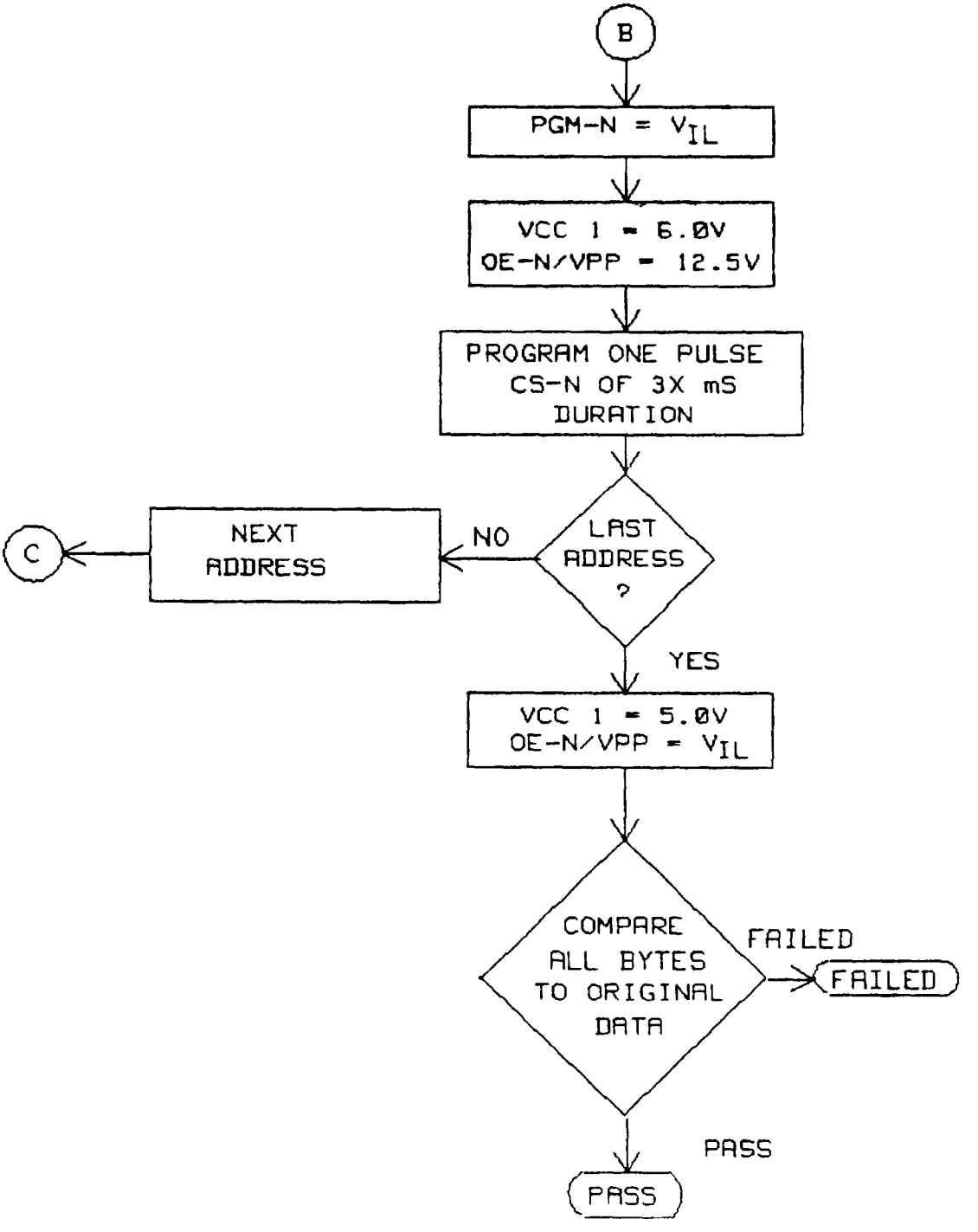


FIGURE 1b. Programming algorithm for UV EPROM.

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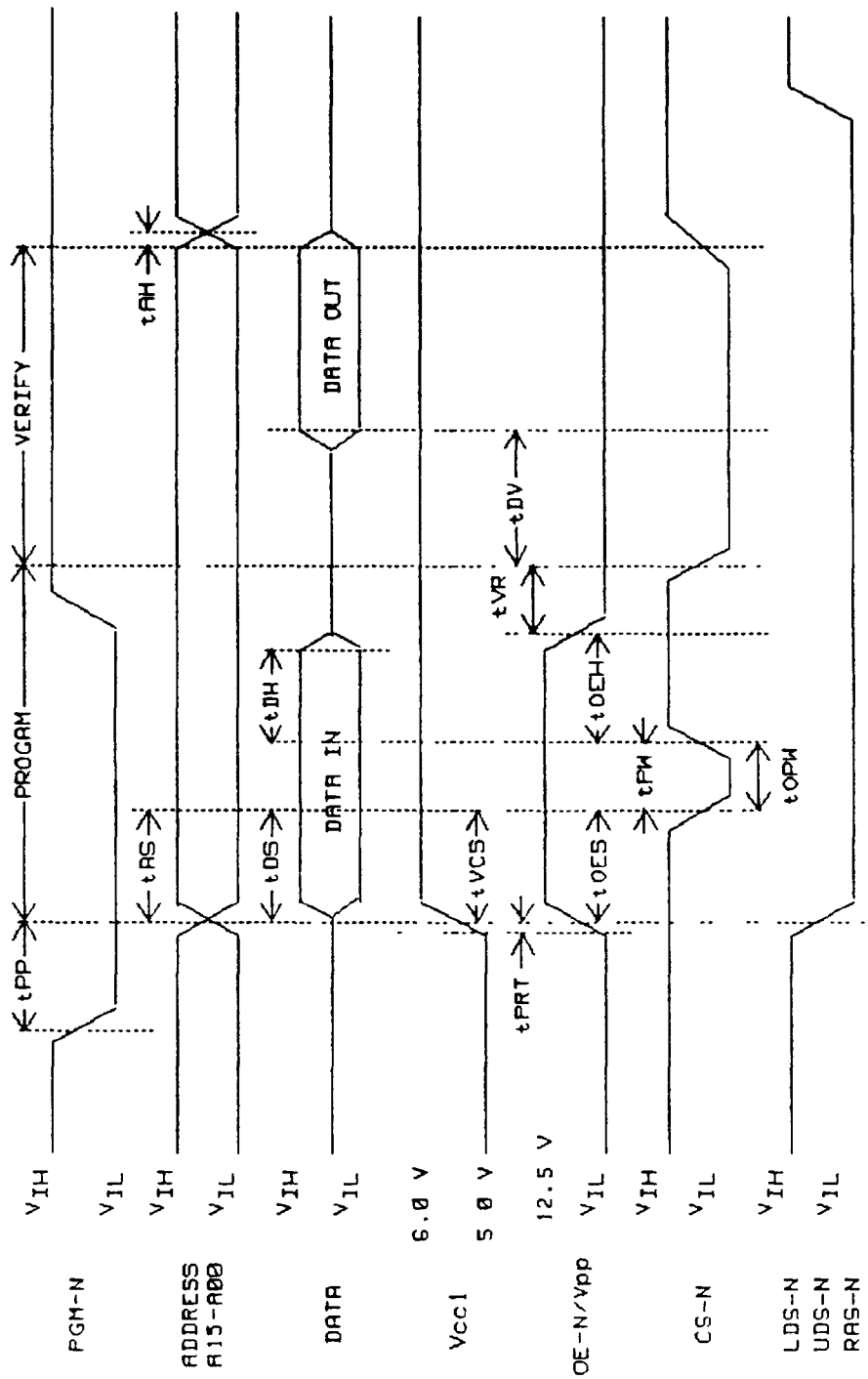


FIGURE 2. Programming timing characteristics for UV EPROM.

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APPENDIX

TABLE I. Programming timing characteristics for UV EPROM.

Symbol	Parameter	Min	Max	Units
tPP	Transceiver direction	2.0	-	uS
tAS	Address setup time	2.0	-	uS
tAH	Address hold time	0.0	-	uS
tDS	Data setup time	2.0	-	uS
tDH	Data hold time	2.0	-	uS
tVCS	Vcc1 setup time	2.0	-	uS
tVR	OE-N/Vpp recovery time	2.0	-	uS
tDV	Data valid from CSLO or CSHI	1.0	-	uS
tOES	OE-N/Vpp setup time	2.0	-	uS
tOEH	OE-N/Vpp hold time	2.0	-	uS
tPRT	OE-N/Vpp pulse rise time	50.0	-	nS
tPW	CS-N initial program pulse width	0.95	1.05	mS
tOPW	CS-N over program pulse width	2.85	78.75	mS

NOTES:

1. T = 25 $\pm$ 5 °C, Vcc1 = 6.0 $\pm$ 0.25 V, OE-N/Vpp = 12.5 $\pm$ 0.5 V.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-W-28787/458	2. DOCUMENT DATE (YYMMDD) 1990 October 26

3. DOCUMENT TITLE
Modules, Digital, Standard Electronic Ultraviolet Light Erasable Programmable Read Only /Memory (UV EPROM) Module

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

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